INDIANA – Verifying (Random) Probing Security through Indistinguishability Analysis
– Extended Version –

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Abstract. While masking is a widely used defense against passive side-channel attacks, its secure implementation in hardware continues to be a manual, complex, and error-prone process.
This paper introduces INDIANA, a comprehensive security verification methodology for hardware masking. Our results include a hardware verification tool, enabling a complete analysis of simulation-based security in the glitch-extended probing model and intra-cycle estimations for leakage probabilities in the random probing model. Notably, INDIANA is the first framework to analyze arbitrary masked circuits in both models, even at the scale of full SPN cipher rounds (e.g., AES), while delivering exact verification results. To achieve accurate and comprehensive verification, we propose a partitionable probing distinguisher that allows for fast validation of probe tuples, surpassing current methods that rely on statistical independence. Furthermore, our approach naturally supports extensions to the random probing model by utilizing Fast Fourier-Hadamard Transformations (FHTs).
Benchmark results show that INDIANA competes effectively with leading probing model verification tools, such as ironMask, maskVerif, and VERICA. INDIANA is also the first tool that is capable to provide intra-cycle estimations of random probing leakage probabilities for large-scale masked circuits.

Keywords: Indistinguishability Analysis · Side-Channel Analysis · Probing Security · Random Probing Security · Security Verification

1 Introduction

With the discovery of Side-Channel Analysis (SCA) [36, 37], security experts have become aware that hardware implementations may inadvertently disclose sensitive information while executing cryptographic algorithms.

In response, *masking* has been extensively applied in practice as a sound and effective countermeasure but its correct and secure implementation in hardware circuits continues to pose a significant challenge. In particular, the abundance of proposed schemes [18, 29–33, 35, 44, 46], along with a notable portion proven to be vulnerable [41], bears witness to the fact that design and implementation of hardware masking schemes still remains a complex and error-prone process.

To address this challenge in a systematic approach, formal definitions of adversary models [6, 21, 32, 45], security properties [4, 5, 19, 40], and architectural conditions [6, 22] are essential. Adhering to these formal criteria is crucial for simplifying, expediting, and automating the assessment of implementation correctness and security.

Consequently, to assess the security of hardware circuits, particularly in the context of masked implementations and passive implementation attacks, these criteria are generally formalized and systematized in terms of leakage models. In this context, a widely accepted model is the d-probing model, introduced by Ishai, Sahai, and Wagner in 2003 [32], where the attacker's view on leakage is represented by the choice and exact knowledge of d intermediate variables. This model was later complemented by Faust et *al.* [22] with consideration of unintentional physical effects naturally occurring in modern CMOS technology, e.g., glitches, transitions, and coupling effects.

Although probing-based leakage models have been shown to facilitate and accelerate security reasoning, they occasionally fall short in accurately representing the intricacies of hardware circuits. Notably, these models neglect certain aspects, such as horizontal attacks [8], which specifically exploit the repeated manipulation of variables within iterative implementations. As a result, the research community is increasingly leaning towards more realistic leakage models, such as the random probing model [21, 32] where leakage is assumed to capture the exact value carried by each wire of the circuit with a probability p. Moreover, this model is intricately linked to the security in the most realistic noisy leakage model [20], where each variable leaks a noisy function of its value.

As leakage models continually advance and security proofs become more complex, there is a growing interest within the community in automating formal security verification. As a result, the development and implementation of automated security reasoning tools in the realm of hardware masking has emerged as a recent and evolving area of research [2, 3, 11, 12, 18, 27, 34, 47].

As a pioneering work, rebecca [12] employs Fourier coefficient estimation to verify standard and glitch-extended probing models. Its successor, COCO [27], extends this methodology to verify masked software running on processors. Shortly after, maskVerif [3] was introduced as an efficient probing security tool; however, its incomplete verification may result in false negatives. Additionally, scVerif [7] extends maskVerif by incorporating customizable leakage models. SILVER [34], the first tool to ensure comprehensive verification, however, is limited by constraints in circuit size and security order. VERICA [47] is an extension of SILVER with Fault Injection Analysis (FIA) support, which allows combined analysis for the first time, but maintains similar complexity limitations. Most recently,

PROLEAD [42] applies logic simulation for probing-based leakage assessment, but can only estimate the security level with a certain confidence.

Expanding the scope of security verification research to the random probing model, VRAPS [9] specifically addresses random probing security, although it is notably limited in terms of efficiency. Similarly, STRAPS [17] adopts the rules from maskVerif to quantify the random probing security, but inherits the constraints of the latter tool. The latest contribution in this domain, ironMask [11], stands out as a high-performing and complete tool, but is strictly constrained to small masking circuits with specific structure, i.e., only supports verification for linear randomness and quadratic gadgets.

Regarding relevant prior research, it is clear that these tools either depend on estimates and heuristics, leading to incomplete results, or they impose significant limitations on the structure and size of the masked circuits they can process. In addition to this, all existing tools also have limited capabilities in accommodating more realistic leakage models, such as those considering glitch-extended d-probing or the random probing leakage model. As a result, the open research challenge is to develop sound, accurate and efficient tools for verifying the security of arbitrary and complex masked hardware circuits under more realistic leakage models (e.g., considerations of glitches or random probing leaks) to overcome the existing limitations.

Contributions. In this work, we introduce a comprehensive and powerful automated verification methodology designed for verifying security of masked hardware. We further instantiate the methodology in terms of a automated verification tool to facilitate effective security validation for large-scale digital circuits, accommodating both the glitch-extended and random probing security models. To outline, our primary contributions can be succinctly described as follows:

- We formalize the prevailing d-probing security concepts by expressing them in terms of the *indistinguishability* of secret-dependent probability distributions observed by adversaries. This formalization is specifically designed for hardware-oriented masking verification, addressing and accommodating the glitch-extended robust d-probing model. In essence, commencing with secret-dependent input distributions and the gate-level netlist of a masked circuit, these distributions continuously transition through the circuit. This process makes it possible to apply the verification procedure iteratively, generating discrete multi-variate probability distributions for adversarial observations (i.e., intermediate wires). The circuit is verified secure if an attacker cannot differentiate all secret-dependent distributions for all observations.
- We extend our primary d-probing security distinguisher to accommodate the more realistic random probing security model by employing Fast Fourier-Hadamard Transformations (FHTs). This extension, in particular, enables the conversion of differences in observed secret-dependent probability distributions into corresponding sets of probes responsible for the information leakage. To elaborate, the utilization of the FHT yields a succinct representa-

tion of all probe tuples contributing to the leak, accelerating the calculation of leakage probabilities in the context of random probing security.
- While the FHT provides the theoretical basis for our approach, the use of Multi-Terminal Binary Decision Diagrams (MTBDDs) as an efficient data structure is the key to the practical implementation of our theoretical concepts. In particular, the structure of the analyzed functions (often) facilitates an efficient mapping to MTBDDs allowing us to practically verify larger probing sets (e.g., for probing sets of $m > 80$ probes) in adequate time although the general complexity of the FHT is still about $\mathcal{O}(m \cdot 2^m)$.
- We present INDIANA, a complete and versatile verification framework designed for the rapid validation of large-scale masked hardware circuits within the glitch-extended probing model. Furthermore, INDIANA facilitates intra-cycle estimation of leakage probabilities in the random probing model for arbitrary large-scale circuits (and exact computation for small-scale designs). We showcase the performance and capabilities of our novel verification tool through comparison to state-of-the-art competitors and various case studies, including the instantiation of a full AES round (i.e., 16 masked S-boxes in parallel) with shared data and key inputs. This results in the (symbolic) simulation of 2^{256} distinct secrets (resulting in 2^{512} different shared input combinations). Notably, INDIANA achieves exhaustive verification in the glitch-extended probing model within less than 45 minutes for the full round of AES, while the intra-cycle estimation of leakage probabilities in the random probing model terminates in less than 4 hours.

Outline. In Section 2, we present the definitions for threshold d-probing security and random probing security based on *indistinguishability* and introduce an efficient leakage detection using the FHT. In Section 3, we then discuss approaches and optimizations for the practical implementation of security verification. In particular, we use novel techniques based on MTBDDs to practically verify the indistinguishability-based definitions and perform leakage detection in the random probing model. Eventually, Section 4 presents several case studies to empirically demonstrate the capabilities and limitations of our verification tool INDIANA, before the paper concludes in Section 5.

Notation. Our basic notation used throughout this work is summarized in Table 1. In general, we write functions in sans serif font (e.g., F) and sets in upper-case characters in a calligraphic font (e.g., $\mathcal{S}$). For a set $\mathcal{S}$, we denote by $2^{\mathcal{S}}$ its power set, i.e., the set of all subsets of $\mathcal{S}$.

2 Side-Channel Security

In SCA an adversary exploits the correlation of physical characteristics, such as timing behavior [36], instantaneous power consumption [37], or electromagnetic emanations [26], with a secret-dependent internal state to gain knowledge

Table 1: Notations used throughout this work.

	Notation	Description
Circuit	n_i	Number of unshared inputs to a circuit
	n_r	Number of randomness inputs to a (masked) circuit
	n_o	Number of unshared outputs of a circuit
	C	Digital logic circuit
	$\mathcal{W}$	Set of wires
SCA	s	Number of shares used by a masking scheme
	d	Security order of a masking scheme (countermeasure)
	$\mathcal{P}$	Set of probes
	Enc	Encoding function for masking
	$\mathsf{A_C}$	Access function to circuit C
	Ex	Probe-extension function
	$\mathcal{L}()$	Adversarial observation (leakage)
	`ProbeSelect`()	Random variable for probe selection

of a secret. The physical reality of such attacks is complex and often highly stochastic. Therefore, the research community has focused on the development and analysis of *security models* that abstract reality to its core component to enable formal reasoning about SCA security. In the following section, we describe the formal context of this work. In particular, we formally define our model for hardware circuits and describe *masking* as a popular countermeasure against SCA. Then, we define the well-known threshold d-probing and random probing model [32] based on indistinguishability of secret-dependent distributions, which is equivalent to the standard simulation-based definitions found in literature.

2.1 Circuit Model

We model a circuit as a Direct Acyclic Graph (DAG) $\mathsf{C} = (\mathcal{G}, \mathcal{W})$, where each vertex (or node) $g \in \mathcal{G}$ represents a gate, an input, or an output of the circuit and each edge $w \in \mathcal{W} = \{w_0, \ldots, w_{|\mathcal{W}|-1}\}$ a wire that carries a value from $\mathbb{F}_2$ and connects two gates. Without loss of generality, we restrict the set of logical gates to $\mathcal{G}_\mathsf{c} = \{\mathsf{xor}, \mathsf{or}, \mathsf{and}, \mathsf{nand}, \mathsf{xnor}, \mathsf{nor}, \mathsf{inv}\}$, which each implement the respective Boolean function. We further restrict the set of memory gates to $\mathcal{G}_\mathsf{m} = \{\mathsf{reg}\}$, where reg represents a clocked register. Let $\mathcal{G}_\mathsf{all}$ be $\mathcal{G}_\mathsf{c} \cup \mathcal{G}_\mathsf{m}$. For convenience, we sometimes assume $\mathcal{W}$ to be an index set, where each wire $w_i \in \mathcal{W}$ is labeled by its index i, i.e., we will use $\mathcal{W} = \{w_0, \ldots, w_{|\mathcal{W}|-1}\}$ and $\mathcal{W} = \{0, \ldots, |\mathcal{W}| - 1\}$ interchangeably.

Definition 1 (Circuit). *A* circuit *with n_i inputs and n_o outputs is a DAG $\mathsf{C} = (\mathcal{G}, \mathcal{W})$, where each vertex takes a label from $\{x_1, \ldots, x_{n_i}, y_1, \ldots, y_{n_o}\} \cup \mathcal{G}_\mathsf{all}$, and which fulfills the following properties:*

1. *For each $j \in \{1, \ldots, n_i\}$ there is exactly one vertex labeled x_j and for each $k \in \{1, \ldots, n_o\}$ there is exactly one vertex labeled y_k.*

2. *Each vertex with a label x_j has in-degree* 0 *and each vertex with label y_k has in-degree* 1 *and out-degree* 0.
3. *Each vertex with a label in* $\{\mathsf{xor}, \mathsf{or}, \mathsf{and}, \mathsf{nand}, \mathsf{xnor}, \mathsf{nor}\}$ *has in-degree* 2.
4. *Each vertex with label* inv *or* reg *has in-degree* 1.

A circuit is a representation of a function $\mathsf{F}\colon \mathbb{F}_2^{n_i} \to \mathbb{F}_2^{n_o}$, that is computed by going iteratively through the gates and assign each wire with the result of the respective logical gate given the values of the input wires. In this sense, each wire $w \in \mathcal{W}$ computes a function $\mathsf{F}_w\colon \mathbb{F}_2^{n_i} \to \mathbb{F}_2$ that is computed by considering the subgraph with the leave w. This motivates the definition of an access function, where we consider each wire $w \in \mathcal{W}$ to be connected to an output node. Hence, intuitively this function provides access to all internal values of C.

Definition 2 (Access to a Circuit). *Given a circuit $\mathsf{C} = (\mathcal{G}, \mathcal{W})$ that implements a function $\mathsf{F}\colon \mathbb{F}_2^{n_i} \to \mathbb{F}_2^{n_o}$. The* access to C *is a function $\mathsf{A}_\mathsf{C}\colon \mathbb{F}_2^{n_i} \to \mathbb{F}_2^{|\mathcal{W}|}$ where the coordinates of A_C correspond to all the edges in C, i.e., for $x \in \mathbb{F}_2^{n_i}$, we have $\mathsf{A}_\mathsf{C}(x) = (w_0, \ldots, w_{|\mathcal{W}|-1})$, where $w_i \in \mathbb{F}_2$ takes the value of the i-th wire in C when the input nodes of C are instantiated with x.*

Note that the notion of the access to a circuit depends on the labeling of the wires. Throughout this work, we assume lexicographic ordering. Further, for a function F mapping to $\mathbb{F}_2^{n_o}$ and a subset $\mathcal{I} \subseteq \{0, \ldots n_o - 1\}$ of output indices, we denote by $\mathsf{F}|_\mathcal{I} = (\mathsf{F}_i)_{i \in \mathcal{I}}$ the subfunction of F that consists of the coordinates of $\mathcal{I}$, mapping to $\mathbb{F}_2^{|\mathcal{I}|}$. Again, we assume that the coordinates of $\mathsf{F}|_\mathcal{I}$ are ordered by the lexicographic ordering of the indices in $\mathcal{I}$.

2.2 Security Mechanism: Masking

One widely-studied countermeasure against SCA is *Boolean masking* [20], where every secret $x_i \in \mathbb{F}_2$ is replaced by a vector $\mathsf{Sh}(x) = \langle x_{i,0}, \ldots x_{i,s-1} \rangle \in \mathbb{F}_2^s$ such that each subset $\hat{\mathcal{X}}$ of $\{x_{i,j} \mid j \in [0, s-1]\}$ with $|\hat{\mathcal{X}}| < s$ is independent of $x_i = \bigoplus_{j=0}^{s-1} x_{i,j}$. We call $x_{i,j}$ a *share* of x_i with *share index* j.

Similarly, a circuit is transformed into a *shared circuit* by transforming each gate into a set of gates that operate on shared values [32]. Specifically, every circuit $\tilde{\mathsf{C}}$, with n_i inputs and n_o outputs, is split into three circuits with the following properties:

$\mathsf{Enc}\colon \mathbb{F}_2^{n_i} \times \mathbb{F}_2^{(s-1)n_i} \to \mathbb{F}_2^{n_i \cdot s}$: A circuit implementing a function that takes as input the n_i inputs to the original circuit $x_1 \ldots x_{n_i}$ and $(s-1)n_i$ random bits $r \in \mathbb{F}_2^{(s-1)n_i}$ and outputs a valid sharing of the input, i.e., for all $i \leq n_i$ an element from $\mathbb{F}_2^s$ with the above property[1] towards the input x_i.

$\mathsf{C}\colon \mathbb{F}_2^{n_i \cdot s} \times \mathbb{F}_2^{n_r} \to \mathbb{F}_2^{n_o \cdot s}$: A masked version of the original circuit, that takes as input the shared values of the inputs and n_r random bits and produces a valid sharing of the output, i.e., for all $j \leq n_o$ an element of $\mathbb{F}_2^s$ with the above property towards the output y_j.

[1] Due to its practical relevance, we focus on Boolean masking here, but Enc can easily be extended to any other masking scheme.

$\mathsf{Dec}\colon \mathbb{F}_2^{n_o \cdot s} \to \mathbb{F}_2^{n_o}$: A circuit that implements a function that given a valid sharing of n_o values produces the unshared values by summing up the shares, i.e., for all $i \leq n_o$ computes $y_i = \bigoplus_{j=0}^{s-1} y_{i,j}$.

For the correctness of the masked circuit, we require that for all inputs $x \in \mathbb{F}_2^{n_i}$, $r_0 \in \mathbb{F}_2^{(s-1)n_i}$, and $r_1 \in \mathbb{F}_2^{n_r}$ it holds that $\tilde{\mathsf{C}}(x) = \mathsf{Dec}(\mathsf{C}(\mathsf{Enc}(x, r_0), r_1))$. Note, if we compute the access A_C of the masked circuit C, we consider only the wires in C, not those in Enc or Dec. In addition, we define the composition $\mathsf{H} \coloneqq \mathsf{A}_\mathsf{C} \circ \mathsf{Enc}$ to be a function mapping from $\mathbb{F}_2^{n_i} \times \mathbb{F}_2^{(s-1)n_i + n_r}$ to $\mathbb{F}_2^{|\mathcal{W}|}$ such that $\mathsf{H}(x, r_0 \| r_1) = \mathsf{A}_\mathsf{C}(\mathsf{Enc}(x, r_0), r_1)$ for all $x \in \mathbb{F}_2^{n_i}$, $r_0 \in \mathbb{F}_2^{(s-1)n_i}$, and $r_1 \in \mathbb{F}_2^{n_r}$.

2.3 Probing Security

A well-established model to argue about side-channel security of masked circuits is the *d-probing model* [32]. Here, an adversary $\mathcal{A}_p$ gets access to a circuit C that can be invoked multiple times. Prior to each invocation, $\mathcal{A}_p$ can select a set $\mathcal{P} \subseteq \mathcal{W}$ of up to d wires to be probed. In the *standard* d-probing model, the view of $\mathcal{A}_p$ on each invocation is defined by the exact values of the probed wires. More specifically, we can express the view of $\mathcal{A}_p$ for a set of probes $\mathcal{P}$ by a subfunction of the encoded access to the circuit C, i.e., by $(\mathsf{A}_\mathsf{C} \circ \mathsf{Enc})|_\mathcal{P}$.

However, in hardware effects such as glitches, transitions, or couplings can cause additional leakage and need to be considered in the security model. For this, the *robust* d-probing model was introduced [22] that extends the set of probes $\mathcal{P}$ by additional probes to capture worst-case assumptions of physical defaults. For example, glitches are temporary and unintentional values in a combinational circuit, caused by timing differences in the computation paths. In the robust d-probing model, those are represented by giving $\mathcal{A}_p$ access to the exact values of the stable inputs to the probes, i.e., registers or primary inputs with no register on the path to the probed wires. To formally capture physical defaults, we define a probe-extension function $\mathsf{Ex}(\mathcal{P})$ for a subset of wires $\mathcal{P} \subseteq \mathcal{W}$ that produces another subset of wires $\mathcal{P}' \subseteq \mathcal{W}$. This allows us to model any additional leakage that is a function of the probed wires, which contains (but is not restricted to) the probe extensions defined in the robust d-probing model. With this, we again express the view of the adversary by a subfunction of the access of the circuit C, namely $(\mathsf{A}_\mathsf{C} \circ \mathsf{Enc})|_{\mathsf{Ex}(\mathcal{P})}$. Note, that for Ex being the identity function id, this results in the standard definition of the d-probing model.

In this context, a circuit C is d-probing secure if any set $\mathcal{P}$ of up to d probes is statistically independent of the secret input [32, 34]. For this work, we reformulate the definition based on indistinguishability. In short, two events $\mathcal{X}$ and $\mathcal{Y}$ are indistinguishable if the two underlying distributions of $\mathcal{X}$ and $\mathcal{Y}$ are equal. To formally capture the definition of probing security, we introduce a leakage function based on indistinguishability.

Definition 3 (Leakage). *Let* $\mathsf{Enc}\colon \mathbb{F}_2^{n_i} \times \mathbb{F}_2^{(s-1)n_i} \to \mathbb{F}_2^{n_i \cdot s}$ *be an encoding and* $\mathsf{A}_\mathsf{C}\colon \mathbb{F}_2^{n_i \cdot s} \times \mathbb{F}_2^{n_r} \to \mathbb{F}_2^{|\mathcal{W}|}$ *be the access to a masked circuit* C *implementing a*

function $\mathsf{F}\colon \mathbb{F}_2^{n_i \cdot s} \times \mathbb{F}_2^{n_r} \to \mathbb{F}_2^{n_o \cdot s}$. *Let* $\mathsf{H} \coloneqq \mathsf{A_C} \circ \mathsf{Enc}$, *which is a function from* $\mathbb{F}_2^{n_i} \times \mathbb{F}_2^{(s-1)n_i+n_r}$ *to* $\mathbb{F}_2^{|\mathcal{W}|}$. *We define*

$$\mathcal{L}_{(\mathsf{Enc},\mathsf{A_C},\mathsf{Ex})}\colon 2^{\mathcal{W}} \to \{0,1\}$$
$$\mathcal{P} \mapsto \begin{cases} 0, \textit{if } \forall x \in \mathbb{F}_2^{n_i}, y \in \mathbb{F}_2^{|\mathsf{Ex}(\mathcal{P})|}\colon \mathsf{Pr}_r[\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})}(x,r) = y] = \mathsf{Pr}_r[\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})}(0,r) = y] \\ 1, \textit{else} \end{cases}$$

to be the leakage of the probeset $\mathcal{P}$ *in the masked circuit* C, *where the probabilities are taken over uniformly random choices of* $r \in \mathbb{F}_2^{(s-1)n_i+n_r}$.

Then, for probing security, we require that there is no leakage regardless of the probe position, for a probe cardinality at most d.

Definition 4 (Probing Security - Indistinguishability). *Let* $\mathsf{Enc}\colon \mathbb{F}_2^{n_i} \times \mathbb{F}_2^{(s-1)n_i} \to \mathbb{F}_2^{n_i \cdot s}$ *be an encoding and* $\mathsf{C} = (\mathcal{G}, \mathcal{W})$ *be a circuit implementing a function* $\mathsf{F}\colon \mathbb{F}_2^{n_i \cdot s} \times \mathbb{F}_2^{n_r} \to \mathbb{F}_2^{n_o \cdot s}$. *We say that* C *is* d*-probing secure with respect to* Enc *and a probe extension* Ex *if the access* $\mathsf{A_C}$ *to* C *fulfills* $\mathcal{L}_{(\mathsf{Enc},\mathsf{A_C},\mathsf{Ex})}(\mathcal{P}) = 0$ *for all sets* $\mathcal{P} \subseteq \mathcal{W}$ *of up to* d *probes.*

Intuitively, the definition for d-probing security fixes the view of $\mathcal{A}_p$ (by fixing the set of probes $\mathcal{P}$) and then requires that $\mathcal{A}_p$ cannot distinguish between different input values $x \in \mathbb{F}_2^{n_i}$. It is easy to see that this is equivalent to the traditional definition requiring statistical independence between the set of probes $\mathcal{P}$ and the input x [34]. See Appendix A for more details.

In a similar way, we can define known notions of composition for probing security (e.g., from [4,5,19]). For this, we fix a subset of input shares and require that the specific value of all other input shares cannot be distinguished from the zero input, i.e., we perform the analysis on the encoded secret $\mathsf{Enc}(x) \in \mathbb{F}_2^{n_i \cdot s}$ instead of the secret $x \in \mathbb{F}_2^{n_i}$. This effectively restricts the input shares the adversary can learn from the probe observation, as usually done by notions of composition. We provide more details in Appendix B but focus on the verification of probing security only for the remainder of this work.

2.4 Random Probing Security

Another established security model for SCA is the *random probing model* [32]. Here, instead of letting the adversary choose a bounded set of probes, an unbounded set of probes is randomly selected and given to the adversary, such that each wire $w \in \mathcal{W}$ is in the set of probes with the same probability p. For this, let $\texttt{ProbeSelect}(\mathsf{C}, p)$, cf. $\mathsf{LeakingWires}$ in [9], be a random variable that, given a circuit C and a probability p, probabilistically chooses a set of probes $\mathcal{P}$, such that each wire $w \in \mathcal{W}$ is probed with probability p independent of all other wires, i.e., $\mathsf{Pr}[w \in \mathcal{P}] = p$. Note, in contrast to the d-probing model, the set $\mathcal{P}$ is not restricted in size. Then, for random probing security, we require the probability that the adversary gets a useful set of probes to be small.

Definition 5 (Random Probing Security - Indistinguishability). *Let* $\mathsf{Enc}\colon \mathbb{F}_2^{n_i} \times \mathbb{F}_2^{(s-1)n_i} \to \mathbb{F}_2^{n_i \cdot s}$ *be an encoding and* $\mathsf{C} = (\mathcal{G}, \mathcal{W})$ *be a circuit implementing a function* $\mathsf{F}\colon \mathbb{F}_2^{n_i \cdot s} \times \mathbb{F}_2^{n_r} \to \mathbb{F}_2^{n_o \cdot s}$*. We say that* C *is* (p, ϵ)*-random probing secure with respect to* Enc *and probe extension* Ex *if the access* A_C *to* C *fulfills*

$$\mathsf{Pr}[\mathcal{L}_{(\mathsf{Enc},\mathsf{A}_\mathsf{C},\mathsf{Ex})}(\mathtt{ProbeSelect}(\mathsf{C}, p)) = 1] \leq \epsilon\,.$$

For the standard definition of the random probing model the probe extension function Ex is selected to be the identity function id, i.e., there is no probe extension. Then, it is easy to see, that this definition is equivalent to the simulation-based definition from Belaïd et al. [9][2], where the simulation of the probes fails with probability ϵ.

2.5 Using FFT for Detecting Leakage

We show how to evaluate the leakage function by computing the Fourier-Hadamard transforms of some associated functions. First, we recall that the *Fourier-Hadamard transform* $\widehat{\mathsf{f}}$ of a function $\mathsf{f}\colon \mathbb{F}_2^m \to \mathbb{R}$ is defined as

$$\widehat{\mathsf{f}}\colon \mathbb{F}_2^m \to \mathbb{R}, \quad \beta \mapsto \sum_{y \in \mathbb{F}_2^m} (-1)^{\langle \beta, y \rangle} \mathsf{f}(y),$$

where $\langle u, v \rangle$ denotes the canonical inner product of the vectors $u, v \in \mathbb{F}_2^m$. For a subset $\mathcal{I}$ of $\{0, 1, \ldots, m-1\}$, we denote by $\mathrm{prec}(\mathcal{I})$ the $|\mathcal{I}|$-dimensional subspace $\{\beta \in \mathbb{F}_2^m \mid \beta_j = 0 \text{ if } j \notin \mathcal{I}\}$ of $\mathbb{F}_2^m$.

Given a function $\mathsf{F}\colon \mathbb{F}_2^{n_i} \times \mathbb{F}_2^{(s-1)n_i + n_r} \to \mathbb{F}_2^{n_o}$ and $x \in \mathbb{F}_2^{n_i}$, we denote by F_x the function obtained from F by fixing x, i.e., $\mathsf{F}_x(z) = \mathsf{F}(x, z)$. For $y \in \mathbb{F}_2^{n_o}$, we denote by $(\mathsf{F})_x^{-1}(y)$ the preimages of y under F_x, i.e., the set of $z \in \mathbb{F}_2^{(s-1)n_i + n_r}$ such that $\mathsf{F}_x(z) = y$. For a given $\mathsf{H}\colon \mathbb{F}_2^{n_i} \times \mathbb{F}_2^{(s-1)n_i + n_r} \to \mathbb{F}_2^{|\mathcal{W}|}$ and $y \in \mathbb{F}_2^{|\mathcal{W}|}$, let us denote $\mathsf{D}_{\mathsf{H},x}(y) \coloneqq |(\mathsf{H})_x^{-1}(y)| - |(\mathsf{H})_0^{-1}(y)| \in \mathbb{R}$, i.e., the difference of the number of preimages of y under H when restricted to input x and the number of premiages of y under H when restricted to input 0. Then, $\mathsf{D}_{\mathsf{H},x}\colon y \mapsto \mathsf{D}_{\mathsf{H},x}(y)$ is a function mapping from $\mathbb{F}_2^{|\mathcal{W}|}$ to $\mathbb{R}$. The main result of this subsection is the following link between the leakage function and the Fourier-Hadamard transform of the functions $\mathsf{D}_{\mathsf{H},x}$. Note that, in the following we identify the wires by their indices, i.e., $\mathcal{W} = \{0, 1, \ldots, |\mathcal{W}| - 1\}$.

Theorem 1. *Let* $\mathsf{Enc}\colon \mathbb{F}_2^{n_i} \times \mathbb{F}_2^{(s-1)n_i} \to \mathbb{F}_2^{n_i \cdot s}$ *be an encoding and* $\mathsf{A}_\mathsf{C}\colon \mathbb{F}_2^{n_i \cdot s} \times \mathbb{F}_2^{n_r} \to \mathbb{F}_2^{|\mathcal{W}|}$ *be the access to a masked circuit* C *implementing a function* $\mathsf{F}\colon \mathbb{F}_2^{n_i \cdot s} \times \mathbb{F}_2^{n_r} \to \mathbb{F}_2^{n_o \cdot s}$*. Let* $\mathsf{H} = \mathsf{A}_\mathsf{C} \circ \mathsf{Enc}$*. Then,* $\mathcal{L}_{(\mathsf{Enc},\mathsf{A}_\mathsf{C},\mathsf{Ex})}(\mathcal{P}) = 0$ *if and only if, for all* $x \in \mathbb{F}_2^{n_i}$*, we have*

$$\forall \beta \in \mathrm{prec}(\mathsf{Ex}(\mathcal{P}))\colon \widehat{\mathsf{D}_{\mathsf{H},x}}(\beta) = 0.$$

[2] Note, however, that [9] considers a different circuit model that includes copy gates instead of wires with arbitrary fan-out. As this may lead to different security levels, we discuss the simulation of copy gates for our tool INDIANA in Section 3.5.

The advantage of this characterization is the fact that the Fourier-Hadamard transform of a function $\mathsf{f}\colon \mathbb{F}_2^m \to \mathbb{R}$ can be computed using $\mathcal{O}(m \cdot 2^m)$ elementary arithmetic operations by using the *Fast Fourier-Hadamard Transform*, see [16, pp.53–54].

For the proof, we use the following two lemmas.

Lemma 1. *Let* $\mathsf{Enc}\colon \mathbb{F}_2^{n_i} \times \mathbb{F}_2^{(s-1)n_i} \to \mathbb{F}_2^{n_i \cdot s}$ *be an encoding and* $\mathsf{A_C}\colon \mathbb{F}_2^{n_i \cdot s} \times \mathbb{F}_2^{n_r} \to \mathbb{F}_2^{|\mathcal{W}|}$ *be the access to a masked circuit* C *implementing a function* $\mathsf{F}\colon \mathbb{F}_2^{n_i \cdot s} \times \mathbb{F}_2^{n_r} \to \mathbb{F}_2^{n_o \cdot s}$*. Then,* $\mathcal{L}_{(\mathsf{Enc},\mathsf{A_C},\mathsf{Ex})}(\mathcal{P}) = 0$ *if and only if*

$$\forall x \in \mathbb{F}_2^{n_i}, y \in \mathbb{F}_2^{|\mathsf{Ex}(\mathcal{P})|}\colon |(\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})})_x^{-1}(y)| = |(\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})})_0^{-1}(y)|.$$

In particular, C *is d-probing secure with respect to* Enc *and* Ex *if and only if, for the access* $\mathsf{A_C}$ *to* C*, we have*

$$\forall \mathcal{P} \in 2^{\mathcal{W}} \text{ with } |\mathcal{P}| \le d\colon \forall x \in \mathbb{F}_2^{n_i}, y \in \mathbb{F}_2^{|\mathsf{Ex}(\mathcal{P})|}\colon |(\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})})_x^{-1}(y)| = |(\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})})_0^{-1}(y)|.$$

Proof. This is an immediate consequence of the fact that

$$\mathsf{Pr}_r[\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})}(x,r) = y] = \frac{|\{r \in \mathbb{F}_2^{(s-1)n_i+n_r} \mid \mathsf{H}|_{\mathsf{Ex}(\mathcal{P})}(x,r) = y\}|}{2^{(s-1)n_i+n_r}} = \frac{|(\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})})_x^{-1}(y)|}{2^{(s-1)n_i+n_r}}.$$

□

Lemma 2. *Let* $\mathsf{f}\colon \mathbb{F}_2^m \to \mathbb{R}$*. Then,* f *is constant and equal to zero if and only if* $\widehat{\mathsf{f}}$ *is constant and equal to zero.*

Proof. This follows directly from the well-known inverse Fourier-Hadamard relation $\widehat{\widehat{\mathsf{f}}} = 2^m \cdot f$, see [16, Cor.4, p.59]. □

Proof (of Thm.1). Let us fix one input $x \in \mathbb{F}_2^{n_i}$. For easier notation, we omit the index H, x in $\mathsf{D}_{\mathsf{H},x}$ and simply write D instead. Let us fix $\mathcal{P} \in 2^{\mathcal{W}}$, which is a subset of $\{0,1,\ldots,|\mathcal{W}|-1\}$.

For $y \in \mathbb{F}_2^{|\mathcal{W}|}$, we denote by $\pi_{\mathsf{Ex}(\mathcal{P})}(y)$ the projection of y to the coordinates indexed by $\mathsf{Ex}(\mathcal{P})$. Then, $\pi_{\mathsf{Ex}(\mathcal{P})}(y) = y'$ for $y' \in \mathbb{F}_2^{|\mathsf{Ex}(\mathcal{P})|}$. Let us define

$$\mathsf{D}_{\mathsf{Ex}(\mathcal{P})}(y') \coloneqq \sum_{y \in \mathbb{F}_2^{|\mathcal{W}|}, \pi_{\mathsf{Ex}(\mathcal{P})}(y) = y'} \mathsf{D}(y). \tag{1}$$

We have $|(\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})})_x^{-1}(y')| - |(\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})})_0^{-1}(y')| = \mathsf{D}_{\mathsf{Ex}(\mathcal{P})}(y')$, hence, by Lemma 1, we have $\mathcal{L}_{(\mathsf{Enc},\mathsf{A_C},\mathsf{Ex})}(\mathcal{P}) = 0$ if and only if $\mathsf{D}_{\mathsf{Ex}(\mathcal{P})}(y') = 0$ for all $y' \in \mathbb{F}_2^{|\mathsf{Ex}(\mathcal{P})|}$ (and all such functions $\mathsf{D}_{\mathsf{Ex}(\mathcal{P})}$ defined by all $x \in \mathbb{F}_2^{n_i}$). Thus, we are going to show that $\mathsf{D}_{\mathsf{Ex}(\mathcal{P})}(y') = 0$ for all $y' \in \mathbb{F}_2^{|\mathsf{Ex}(\mathcal{P})|}$ holds if and only if $\widehat{\mathsf{D}}(\beta) = 0$ holds for all $\beta \in \mathrm{prec}(\mathsf{Ex}(\mathcal{P}))$.

Clearly, the mapping $\beta \mapsto \pi_{\mathsf{Ex}(\mathcal{P})}(\beta)$ is a vector space isomorphism from $\mathrm{prec}(\mathsf{Ex}(\mathcal{P}))$ to $\mathbb{F}_2^{|\mathsf{Ex}(\mathcal{P})|}$. Let $\beta \in \mathrm{prec}(\mathsf{Ex}(\mathcal{P}))$. Then,

$$\begin{aligned}\widehat{\mathsf{D}}(\beta) &= \sum_{y\in\mathbb{F}_2^{|\mathcal{W}|}} (-1)^{\langle\beta,y\rangle}\mathsf{D}(y) = \sum_{y'\in\mathbb{F}_2^{|\mathsf{Ex}(\mathcal{P})|}} \sum_{y\in\mathbb{F}_2^{|\mathcal{W}|},\pi_{\mathsf{Ex}(\mathcal{P})}(y)=y'} (-1)^{\langle\beta,y\rangle}\mathsf{D}(y)\\ &= \sum_{y'\in\mathbb{F}_2^{|\mathsf{Ex}(\mathcal{P})|}} (-1)^{\langle\pi_{\mathsf{Ex}(\mathcal{P})}(\beta),y'\rangle} \sum_{y\in\mathbb{F}_2^{|\mathcal{W}|},\pi_{\mathsf{Ex}(\mathcal{P})}(y)=y'} \mathsf{D}(y)\\ &= \sum_{y'\in\mathbb{F}_2^{|\mathsf{Ex}(\mathcal{P})|}} (-1)^{\langle\pi_{\mathsf{Ex}(\mathcal{P})}(\beta),y'\rangle}\mathsf{D}_{\mathsf{Ex}(\mathcal{P})}(y').\end{aligned}$$

Now, if $\mathsf{D}_{\mathsf{Ex}(\mathcal{P})}(y') = 0$ holds for all $y' \in \mathbb{F}_2^{|\mathsf{Ex}(\mathcal{P})|}$, clearly, by the above equation we have $\widehat{\mathsf{D}}(\beta) = 0$ for all $\beta \in \mathrm{prec}(\mathsf{Ex}(\mathcal{P}))$. For the converse, let us assume that $\widehat{\mathsf{D}}(\beta) = 0$ for all $\beta \in \mathrm{prec}(\mathsf{Ex}(\mathcal{P}))$. By the above equation, $\widehat{\mathsf{D}}(\beta)$ is equal to the Fourier-Hadamard transform at point $\pi_{\mathsf{Ex}(\mathcal{P})}(\beta)$ of the function $\mathsf{D}_{\mathsf{Ex}(\mathcal{P})} \colon \mathbb{F}_2^{|\mathsf{Ex}(\mathcal{P})|} \to \mathbb{R}$. Hence, $\widehat{\mathsf{D}_{\mathsf{Ex}(\mathcal{P})}} = 0$ by assumption. By Lemma 2, it follows that $\mathsf{D}_{\mathsf{Ex}(\mathcal{P})}(y') = 0$ for all $y' \in \mathbb{F}_2^{|\mathsf{Ex}(\mathcal{P})|}$, which concludes the proof. □

Theorem 1 gives us a way to compute the leakage as follows: For $x \in \mathbb{F}_2^{n_i}$, let $\mathsf{g}_{\mathsf{H},x} \colon \mathbb{F}_2^{|\mathcal{W}|} \to \mathbb{F}_2$ be the Boolean function defined by $\mathsf{g}_{\mathsf{H},x}(\beta) = 0$ if and only if $\widehat{\mathsf{D}_{\mathsf{H},x}}(\beta) = 0$ and $\mathsf{g}_{\mathsf{H},x}(\beta) = 1$ otherwise. Then,

$$\mathcal{L}_{(\mathsf{Enc},\mathsf{A_C},\mathsf{Ex})}(\mathcal{P}) = \bigvee_{x\in\mathbb{F}_2^{n_i}} \bigvee_{\beta\in\mathrm{prec}(\mathsf{Ex}(\mathcal{P}))} \mathsf{g}_{\mathsf{H},x}(\beta). \tag{2}$$

This formula is useful to assess the security of a circuit in the random probing model, where we need to analyze $\mathsf{Pr}[\mathcal{L}_{(\mathsf{Enc},\mathsf{A_C},\mathsf{Ex})}(\mathtt{ProbeSelect}(\mathsf{C},p)) = 1]$.

The case of size-preserving probe-extension functions. In the special case where Ex is bijective and preserves the size of its input set (for example when $\mathsf{Ex} = \mathsf{id}$), Theorem 1 implies a link between d-probing security and d-resiliency of the functions $\mathsf{D}_{\mathsf{H},x}$. A function $\mathsf{f} \colon \mathbb{F}_2^m \to \mathbb{R}$ is called *d-resilient*[3] if $\widehat{\mathsf{f}}$ is zero on all inputs with Hamming weight less than or equal to d. We use $\mathrm{wt}(v)$ to denote the Hamming weight of a binary vector v.

Corollary 1. *Let* $\mathsf{Enc} \colon \mathbb{F}_2^{n_i} \times \mathbb{F}_2^{(s-1)n_i} \to \mathbb{F}_2^{n_i\cdot s}$ *be an encoding and* $\mathsf{A_C} \colon \mathbb{F}_2^{n_i\cdot s} \times \mathbb{F}_2^{n_r} \to \mathbb{F}_2^{|\mathcal{W}|}$ *be the access to a masked circuit* C *implementing a function* $\mathsf{F} \colon \mathbb{F}_2^{n_i\cdot s} \times \mathbb{F}_2^{n_r} \to \mathbb{F}_2^{n_o\cdot s}$. *Let* Ex *be a bijective probe-extension function that preserves the size of its input set. Then, the following assertions are equivalent:*

1. C *is d-probing secure with respect to* Enc *and* Ex.

[3] In the literature, d-resiliency is usually defined for a Boolean function $\mathsf{g} \colon \mathbb{F}_2^m \to \mathbb{F}_2$. Then, the usual notion of d-resiliency corresponds to ours if f is the sign function of g, i.e., $\mathsf{f} = (-1)^{\mathsf{g}}$.

2. The functions $\mathsf{D}_{\mathsf{H},x}$ are d-resilient for all $x \in \mathbb{F}_2^{n_i}$.

Proof. By definition, C is d-probing secure with respect to Enc and Ex if and only if $\mathcal{L}_{(\mathsf{Enc},\mathsf{A_C},\mathsf{Ex})}(\mathcal{P}) = 0$ for all $\mathcal{P} \subseteq \mathcal{W}$ with $|\mathcal{P}| \leq d$. If Ex is bijective and preserves the size of its input set, this is equivalent to $\mathcal{L}_{(\mathsf{Enc},\mathsf{A_C},\mathsf{id})}(\mathcal{P}) = 0$ for all $\mathcal{P} \subseteq \mathcal{W}$ with $|\mathcal{P}| \leq d$. The result now follows from Theorem 1 and the fact that the set $\{\beta \in \mathbb{F}_2^{|\mathcal{W}|} \mid \mathrm{wt}(\beta) \leq d\}$ is equal to the union of all sets $\mathrm{prec}(\mathcal{P})$ with $|\mathcal{P}| \leq d$. □

In this case, the maximum d for which a circuit is d-probing secure can be computed by Algorithm 1 given below. Note that in our application, the probe-extension function is more complex, so we do not actually use Algorithm 1. Still, we decided to list it for completeness.

Algorithm 1: Finding the largest d such that C is d-probing secure with respect to Enc and Ex, where Ex is bijective and preserves the size of the input set

```
d ← |W|
for x ∈ F_2^{n_i} do
    Compute D_{H,x}: F_2^{|W|} → R
    Compute D̂_{H,x} from D_{H,x} using FFT        ▷ Complexity |W| · 2^{|W|}
    if max{t | ∀β ∈ F_2^{|W|}, wt(β) ≤ t: D̂_{H,x}(β) = 0} < d then
        d ← max{t | ∀β ∈ F_2^{|W|}, wt(β) ≤ t: D̂_{H,x}(β) = 0}
return d
```

3 Security Verification

This section explains the practical implementation of the formal models and indistinguishability analysis enabling an automated verification of the security definitions outlined in Section 2.

3.1 Preliminaries

To facilitate the efficient representation and manipulation of Boolean functions and probability distributions, we revisit the fundamental concepts and principles underlying Binary Decision Diagrams (BDDs) and MTBDDs.

Binary Decision Diagrams. BDDs are a fundamental data structure in computer science to represent Boolean functions [1,14]. In general, a BDD is a concise and unique (i.e., canonical) graph-based representations of a Boolean function.

Definition 6. *A Reduced Ordered Binary Decision Diagram (ROBDD)*[4] *is a pair* (π, G)*, where* π *denotes a fixed ordering over an variable set* $\mathcal{X} \coloneqq \{x_1, x_2, \ldots, x_n\}$ *and* $\mathsf{G} = (\mathcal{V}, \mathcal{E})$ *represents a rooted DAG with the properties:*

1. *There is a single root and each vertex* $v \in \mathcal{V}$ *is either a non-terminal node or a terminal node. A terminal node is taking a value in the value set* $\mathbb{B} \coloneqq \{0,1\}$*. There is only one terminal node labeled* 0 *and only one terminal node labeled* 1 *(no duplicate terminal nodes).*
2. *Each non-terminal node* v *is labeled with a variable in* $x_i \in \mathcal{X}$*, denoted as* $\mathsf{var}(v)$*, and has exactly two child nodes in* $\mathcal{V}$ *which are denoted as* $\mathsf{then}(v)$ *and* $\mathsf{else}(v)$*.*
3. *For each path from the root node to a terminal node, the variables in* $\mathcal{X}$ *are encountered at most once and in the same order defined by the ordering* π *that is a bijection* $\pi : \{1,2,...,n\} \to \mathcal{X}$*.*
4. *There is no node* $v \in \mathcal{V}$ *such that* $\mathsf{then}(v) = \mathsf{else}(v)$ *and, if* $v, v' \in \mathcal{V}$ *with* $(\mathsf{var}(v), \mathsf{then}(v), \mathsf{else}(v)) = (\mathsf{var}(v'), \mathsf{then}(v'), \mathsf{else}(v'))$*, we have* $v = v'$*.*

Each BDD with single root $v \in \mathcal{V}$ recursively defines a Boolean function $\mathsf{f}_v \colon \mathbb{F}_2^n \to \mathbb{F}_2$ such that, if v is a terminal node $b \in \mathbb{B}$, then $\mathsf{f}_v = b$. Otherwise, if v is a non-terminal node and $\mathsf{var}(v) = x_i$, then f_v is defined by the Shannon decomposition $\mathsf{f}_v = x_i \cdot \mathsf{f}_{\mathsf{then}(v)} + \overline{x_i} \cdot \mathsf{f}_{\mathsf{else}(v)}$. Conversely, any Boolean function can be described by a BDD in this way.

Boolean operations over BDDs. For any BDD, the RESTRICT(f, x, b) operator returns the Shannon co-factor $\mathsf{f}|_{x=b}$ while the If-Then-Else operator $\mathrm{ITE}(\mathsf{f}, \mathsf{g}, \mathsf{h})$ composes the functions $\mathsf{f}, \mathsf{g}, \mathsf{h} \colon \mathbb{F}_2^n \to \mathbb{F}_2$ such that

$$\mathrm{ITE}(\mathsf{f}, \mathsf{g}, \mathsf{h}) = \mathsf{f} \cdot \mathsf{g} + \bar{\mathsf{f}} \cdot \mathsf{h}.$$

The ITE operator can be extended to functions $\mathsf{f}, \mathsf{g}, \mathsf{h}$ with not necessarily equal input domains $\mathbb{F}_2^n$, $\mathbb{F}_2^{n'}$, and $\mathbb{F}_2^{n''}$, respectively, by expanding the function f (resp., g, h) to a function $\tilde{\mathsf{f}} \colon \mathbb{F}_2^{\max\{n,n',n''\}} \to \mathbb{F}_2$ defined by $\tilde{\mathsf{f}}(x_1, \ldots, x_n, \ldots, x_{\max\{n,n',n''\}}) = \mathsf{f}(x_1, \ldots, x_n)$ (defining $\tilde{\mathsf{g}}, \tilde{\mathsf{h}}$ respectively) and defining $\mathrm{ITE}(\mathsf{f}, \mathsf{g}, \mathsf{h}) \coloneqq \mathrm{ITE}(\tilde{\mathsf{f}}, \tilde{\mathsf{g}}, \tilde{\mathsf{h}})$.

Further, any function $\mathsf{f} = \mathsf{f}_{v_1} \star \mathsf{f}_{v_2}$, where $\star$ is an arbitrary binary Boolean operation, can be derived and composed recursively as:

$$\begin{aligned} \mathsf{f} &= x_i \cdot \mathsf{f}|_{x_i=1} + \overline{x_i} \cdot \mathsf{f}|_{x_i=0} \\ &= x_i \cdot (\mathsf{f}_{v_1} \star \mathsf{f}_{v_2})|_{x_i=1} + \overline{x_i} \cdot (\mathsf{f}_{v_1} \star \mathsf{f}_{v_2})|_{x_i=0} \\ &= x_i \cdot (\mathsf{f}_{v_1}|_{x_i=1} \star \mathsf{f}_{v_2}|_{x_i=1}) + \overline{x_i} \cdot (\mathsf{f}_{v_1}|_{x_i=0} \star \mathsf{f}_{v_2}|_{x_i=0}) \end{aligned} \tag{3}$$

Moreover, *existential quantification* with respect to a variable x is a common BDD operation that computes $\exists_x \mathsf{f} \coloneqq \mathsf{f}|_{x=1} + \mathsf{f}|_{x=0}$. If $x_1, \ldots, x_k$ are k variables, the existential quantification $\exists_{x_1,\ldots,x_k} \mathsf{f}$ is recursively defined as $\exists_{x_k} \exists_{x_1,\ldots,x_{k-1}} \mathsf{f}$.

[4] For the sake of simplicity, we use the term BDD instead of ROBDD in the context of this paper.

Multi-Terminal Binary Decision Diagrams. MTBDDs are an extension of BDDs to represent functions from a multi-dimensional Boolean domain to an arbitrary value set $\mathbb{D}$. Specifically, if $\mathbb{D} = \mathbb{B}$ then MTBDDs reduce to BDDs while otherwise they can represent arbitrary functions of type $\mathsf{f} : \mathbb{F}_2^n \to \mathbb{D}$.

Remark. In the remainder of this work, we specifically assume $\mathbb{D} = \mathbb{Z}$ and heavily rely on the *existential quantification* operation over MTBDDs. In this case, the $+$ operation is an arithmetic addition rather than the logic OR operation.

3.2 Fundamental Implementation Assumptions

Before we explain the details of the security verification concept, we briefly summarize essential initial assumptions.

Circuit Representation. Due to the limitation of the circuit model to represent circuits in terms of DAGs, as outlined in Definition 1, it is necessary that any iterative or looped circuit is first transformed into a functionally equivalent circuit by unrolling and pipelining the clocked stages.

Input Distributions. For both considered information leakage models and in accordance with our model, we assume both the initial sharing $r_0 \in \mathbb{F}_2^{(s-1)n_i}$ of secrets and the refreshing $r_1 \in \mathbb{F}_2^{n_r}$ of intermediate computation the application of independent and identically distributed (i.i.d.) random bits.

Extension Function. Glitches are generally known as a major source of information leakage [38, 39] for hardware implementations. Since our verification is primarily focused on hardware-related verification of security properties, we opted for the glitch-extended probing model [22]. In particular, it is assumed that each probe has a worst case extension that additionally leaks all (stable) inputs necessary for the generation of the probed signal.

For the random probing model, however, hardware effects such as glitches have not yet been considered in most of the related literature. Here, we adhere to common practice and therefore only use the identity function id as an extension function within the random probing model, although an extension is generally possible (see [11]) and not inhibited by our methodology.

3.3 Glitch-Extended Threshold Probing Model

Verifying probing security necessitates the exact knowledge of the leakage function (cf. Definition 3). Accordingly, this requires the knowledge of the encoding function Enc, the access function $\mathsf{A_C}$, and potentially the extension function Ex.

Encoding Function. The encoding function is used to achieve a valid and secure sharing of the inputs that are subsequently processed by the masked version of the circuit. In this regard, the type of encoding function is purely determined by the specific masking scheme used to generate the masked circuit.

In our implementation, we have chosen to primarily support Boolean masking due to its prevalent adoption in symmetric cryptography. Nevertheless, it is important to emphasize that our methodology and indistinguishability verification framework are theoretically capable of supporting a wide range of encoding functions, e.g., used for arithmetic or multiplicative masking schemes.

Glitch-extended Access Function. According to Section 2.1, the function $\mathbb{F}_2^{n_i} \times \mathbb{F}_2^{(s-1)n_i} \times \mathbb{F}_2^{n_r} \to \mathbb{F}_2^{m \cdot s}$ of a masked circuit is computed by iteratively assigning each wire with the result of its driving logic gate. In particular, since each wire $w \in \mathcal{W}$ computes a Boolean function $\mathbb{F}_2^{n_i} \times \mathbb{F}_2^{(s-1)n_i} \times \mathbb{F}_2^{n_r} \mapsto \mathbb{F}_2$, this can be generated and stored efficiently with the aid of BDDs. More specifically, each input of the circuit is assigned with a new Boolean variable and the logic gates of the circuit are evaluated iteratively given the BDDs of their inputs. Then, given the set of all BDDs associated with a wire of the circuit, this directly provides the access function of the circuit that can be used to compute $\mathsf{H} = \mathsf{A_C} \circ \mathsf{Enc}$.

For glitch-extended probing security verification, the limited view of an adversary is further captured by a subfunction $\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})}$, with $\mathsf{Ex}(\mathcal{P})$ being the worst-case glitch-extension function as defined before. Hence, in selecting all subsets of wires $\mathcal{P} \subseteq \mathcal{W}$ with $|\mathcal{P}| \leq d$ and generation of the associated extension sets $\mathcal{P}' \subseteq \mathcal{W}$, we can select the corresponding BDDs of the probed wires, to easily express all glitch-extended adversarial views $\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})}$.

Please note, as indicated before, that we have decided not to use the FHT here for two reasons: Firstly, the glitch-extension function is not bijective and size-preserving, and secondly, in practice often only small d (compared to the total number of possible probe locations) are considered, for which a naive enumeration of all possible functions is still feasible so that the application of the FHT shows hardly any advantages.

Nevertheless, in the case of the random probing model, the FHT-based verification has a clear advantage, which is why we use it for the corresponding case studies in Section 4.2.

Leakage Function. According to Lemma 1, knowledge of $|(\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})})_x^{-1}(y)|$ (for all $x \in \mathbb{F}_2^{n_i}$ and all $y \in \mathbb{F}_2^{|\mathsf{Ex}(\mathcal{P})|}$) is sufficient to verify absence of information leakage for any adversarial observation $\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})}$. For this, we introduce our generation methodology, which particularly relies on BDDs and MTBDDs, in the following paragraphs in more detail.

Definition 7 (Transition). *Let* $\mathsf{Enc}\colon \mathbb{F}_2^{n_i} \times \mathbb{F}_2^{(s-1)n_i} \to \mathbb{F}_2^{n_i \cdot s}$ *be an encoding and* $\mathsf{A_C}\colon \mathbb{F}_2^{n_i \cdot s} \times \mathbb{F}_2^{n_r} \to \mathbb{F}_2^{|\mathcal{W}|}$ *be the access to a masked circuit* C *implementing a function* $\mathsf{F}\colon \mathbb{F}_2^{n_i \cdot s} \times \mathbb{F}_2^{n_r} \to \mathbb{F}_2^{n_o \cdot s}$*. Let* $\mathsf{H} \coloneqq \mathsf{A_C} \circ \mathsf{Enc}$*, which is a function from* $\mathbb{F}_2^{n_i} \times \mathbb{F}_2^{(s-1)n_i + n_r}$ *to* $\mathbb{F}_2^{|\mathcal{W}|}$*. We define*

$$\mathcal{T}_{(\mathsf{Enc},\mathsf{A_C},\mathsf{Ex}(\mathcal{P}))}\colon \mathbb{F}_2^{s\cdot n_i+n_r+|\mathsf{Ex}(\mathcal{P})|} \to \{0,1\}$$
$$(x,r,y) \mapsto \begin{cases} 1 & \textit{if } \mathsf{H}|_{\mathsf{Ex}(\mathcal{P})}(x,r) = y \\ 0 & \textit{else} \end{cases}$$

to be the transition function for a tuple (x,r,y) *in the masked circuit* C.

Given this definition and knowledge of $\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})}$, we can directly calculate the transition as follows:

$$\mathcal{T}_{(\mathsf{Enc},\mathsf{A_C},\mathsf{Ex}(\mathcal{P}))}(x,r,y) = \prod_{i=0}^{|\mathsf{Ex}(\mathcal{P})|-1} \overline{(\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})_i}(x,r) \oplus y_i)}$$

where $\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})_i}$ and y_i denote the i-th coordinate function of the adversarial view and the i-th bit of y, respectively.

Based on this, the frequencies $|(\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})})_x^{-1}(y)| = \exists_r \mathcal{T}_{(\mathsf{Enc},\mathsf{A_C},\mathsf{Ex}(\mathcal{P}))}(x,r,y)$ are generated by using the existential quantification operator for MTBDDs with respect to r. Note that this implicitly considers a uniform distribution for $|r|$ (according to our basic implementation assumptions). However, accounting for non-uniform distributions is possible by scaling of the frequencies with respect to the different occurrences of r.

Eventually, indistinguishability of the adversarial observations, according to Definition 4, is implemented and checked with simple subtraction according to $\mathsf{D}_{\mathsf{Ex}(\mathcal{P})}$ (as defined in Eq.(1)) such that $\mathcal{L}_{(\mathsf{Enc},\mathsf{A_C},\mathsf{Ex})}(\mathcal{P}) = 0$ if $\mathsf{D}_{\mathsf{Ex}(\mathcal{P})} = 0$ (for all such functions $\mathsf{D}_{\mathsf{Ex}(\mathcal{P})}$ defined by all $x \in \mathbb{F}_2^{n_i}$) and $\mathcal{L}_{(\mathsf{Enc},\mathsf{A_C},\mathsf{Ex})}(\mathcal{P}) = 1$ otherwise.

3.4 Standard Random Probing Model

Since we assume the identity as an extension function, i.e., $\mathsf{Ex} = \mathsf{id}$ and $\mathcal{P} = \mathcal{W}$ for verification in the standard random probing model, it is possible to efficiently use the FHT to calculate the random probing leakage probabilities. In the following, we therefore describe the efficient calculation of FHT for MTBDDs on the one hand and the straightforward extraction of the leakage probabilities on the other.

The Fast Fourier-Hadamard Transformation. For the FHT, we assume that the function $\mathsf{H}|_{\mathcal{W}} = \mathsf{A_C} \circ \mathsf{Enc}$, considering full access to the masked circuit C, was generated according to the description in the previous section and is provided as MTBDD. In this case, Algorithm 2 describes the exact procedure for generation of the FHT using two fundamental BDD and MTBDD operators RESTRICT and ITE.

In this context, we note that $\mathsf{D}_{\mathsf{H},x}(y) = 0$ for all y is a sufficient but not necessary condition for the absence of leakage. Instead, the sums of $\mathsf{D}_{\mathsf{H},x}$ have to be considered (Equation 1) which can be more efficiently determined using the FHT. As the construction of $\widehat{\mathsf{D}_{\mathsf{H},x}}$ naturally aligns with the Shannon Decomposition

Algorithm 2: Computing the Fast Fourier-Hadamard Transformation using the RESTRICT and ITE MTBDD operations with linear complexity in the number of variables.

$\widehat{\mathsf{D}_{\mathsf{H},x}} \leftarrow \mathsf{D}_{\mathsf{H},x}$
for $x_i \in x$ **do**
  $\mathsf{D}_0 \leftarrow \widehat{\mathsf{D}_{\mathsf{H},x}}|_{x_i=0}$ ▷ RESTRICT
  $\mathsf{D}_1 \leftarrow \widehat{\mathsf{D}_{\mathsf{H},x}}|_{x_i=1}$ ▷ RESTRICT
  $\widehat{\mathsf{D}_{\mathsf{H},x}} \leftarrow \overline{x_i}(\mathsf{D}_0 + \mathsf{D}_1) + x_i(\mathsf{D}_0 - \mathsf{D}_1)$ ▷ ITE
return $\widehat{\mathsf{D}_{\mathsf{H},x}}$

Algorithm 3: Expanding the FHT to the leakage function using RESTRICT MTBDD operator.

$\mathsf{g}_{\mathsf{H},x} \leftarrow \widehat{\mathsf{D}_{\mathsf{H},x}}$
for $\beta_i \in \beta$ **do**
  $\mathsf{g}_0(\beta) \leftarrow \mathsf{g}_{\mathsf{H},x}(\beta)|_{\beta_i=0}$ ▷ RESTRICT
  $\mathsf{g}_{\mathsf{H},x}(\beta) \leftarrow \mathsf{g}_{\mathsf{H},x}(\beta) \vee \mathsf{g}_0(\beta)$
return $\mathsf{g}_{\mathsf{H},x}$

inherent to MTBDDs, it can be efficiently computed using the basic RESTRICT and ITE operations (Algorithm 2), making the data structure of MTBDDs a natural choice. Although the general complexity for the FHT is about $\mathcal{O}(m \cdot 2^m)$, we are still able to handle large m (e.g., $m > 80$) in adequate time due to the structure of the analyzed functions and their efficient mapping to MTBDDs[5].

The Leakage Function. Given $\widehat{\mathsf{D}_{\mathsf{H},x}}(\beta)$ in MTBDD representation, $\mathsf{g}_{\mathsf{H},x}$ can be retrieved rapidly in converting all non-zero terminal nodes to 1, i.e., $\mathsf{g}_{\mathsf{H},x}(\beta) = 0$ if and only if $\widehat{\mathsf{D}_{\mathsf{H},x}}(\beta) = 0$ and $\mathsf{g}_{\mathsf{H},x}(\beta) = 1$ otherwise. Then, $\bigvee_{\beta \in \mathrm{prec}(\mathsf{Ex}(\mathcal{P}))} \mathsf{g}_{\mathsf{H},x}(\beta)$ is derived according to Algorithm 3 (using the RESTRICT operator). The correctness and completeness directly follows from the iterative accumulation of all restricted functions $\mathsf{g}_0(\beta)$, i.e., for all values of β. In particular, for all β where $\mathsf{g}_{\mathsf{H},x}(\beta)|_{\beta_i=0} = 1$, it holds that $\mathsf{g}_{\mathsf{H},x}(\beta) \vee \mathsf{g}_{\mathsf{H},x}(\beta)|_{\beta_i=0} = 1$ for $\beta_i \in \{0,1\}$.

Eventually, computation of the random probing leakage probability requires a slightly modified SATCOUNT operator, according to Algorithm 4.

[5] A clear estimate of complexity is difficult, as it depends on the graph structure and order of the variables in the MTBDDs. However, finding a compact MTBDD structure by means of (dynamic) variable reordering is an NP-complete problem [13], which accordingly emphasizes the heuristic character (depending on the selected variable sorting) of our approach.

Algorithm 4: Modified SATCOUNT BDD operator to compute the leakage probability for a leakage function f and a leakage probability p.

```
1 Function LeakageProbability(f, p):
2     if f = 0 then
3         return 0
4     if f = 1 then
5         return 1
6     Pick next x_i ∈ x
7     p_0 ←LeakageProbability(f|_{x_i=0}, p)        ▷ RESTRICT
8     p_1 ←LeakageProbability(f|_{x_i=1}, p)        ▷ RESTRICT
9     return p_0 + p · (p_1 − p_0)
```

3.5 Additional Implementation Optimizations

To enable practical verification, especially for the targeted large-scale circuits, we discuss additional implementation optimizations in the following paragraphs that simplify and accelerate execution of our verification tool.

Problem-space Partitioning. A classic approach in computer science for handling large problems and search spaces is to break the problem down into smaller sub-problems that can be examined independently of each other before the partial results are finally combined to form the overall result. In this context, we would like to discuss in particular the partitioning of the circuit in width and depth respectively spatial and temporal direction.

Space (circuit breadth). First, it can be observed that for independent modules and components that process independent parts of the secret, the adversarial observations are also independent and can only leak information about the processed part of the secret. Accordingly, these parts can be verified independently and only then the results can be combined. For this purpose, we have implemented an initial information flow analysis for partitioning the circuit with respect to the processed parts of the secret. The examination of smaller sub-components in particular can considerably accelerate the overall verification process without the results losing their significance and precision.

Time (circuit depth). Leveraging the fact that we only consider unrolled and pipelined circuits, we can also partition the verification effort in the temporal dimension of the circuit. Specifically, limiting the verification results to only intra-cycle accuracy, i.e., only considering probes placed within a single cycle respectively pipelining stage, we can partition the computational effort of generating the adversarial view $\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})}$.

In fact, this approach is based on the implementation of a function that allows generating the appropriate output distribution for a given input distribution and a corresponding transition function. Given this, we can compute the

distribution of the adversarial view for each cycle individually in first generating the input distributions of each cycle iteratively as the output distribution of the previous cycle and then leveraging the same function to compute the adversarial view. This approach particularly limits the complexity of the intermediate transition functions in order to improve the evaluation performance. Note, that this decision is also based on the assumption that common practical leakage assessment methodologies (e.g., Test Vector Leakage Assessment (TVLA)) for hardware platforms mostly provide univariate test results.

Parallelism. In particular, the decomposition of the problem space into independent sub-problems favors the parallelization of processing, especially for modern multi-core processor architectures. Accordingly, the different partitions of the individual stages are processed in parallel on independent processor cores, depending on availability, so that the execution time of our verification can continue to be significantly accelerated.

Approximation. Another option for optimizing the tool runtime concerns the accuracy of the verification results. This is particularly interesting for the random probing model, as the combinatorial complexity increases exponentially in the circuit size due to the complete enumeration of all probe combinations, so that even the benefits of partitioning quickly reach their limits.

In this case, we suggest that both the maximum number of probes and the number of probe combinations taken into account can be limited by the user. However, this means that not all combinations are enumerated and considered. As a result, the tool no longer reports an exact leakage probability, but instead determines a lower and an upper bound for the leakage probability. In particular, it is conservatively assumed for the lower bound that all unconsidered combinations are secure, whereas the exact opposite is assumed for the upper bound, i.e., that all unconsidered combinations are insecure (cf. [9]).

By varying the number of samples, the runtime and memory requirements can be adjusted, but at the expense of the accuracy of the verification results.

Simulation of Copy Gates. In order to simulate a potential amplification effect for information leakage due to repeated loading of values, a circuit model was proposed in [9] that introduces so-called copy gates.

Using a copy gate, an input value is copied to two output values. For instance, for a wire with fan-out n, this software-oriented model would have to take $\nu = 2n-1$ copies into account. However, a hardware-oriented model would instead only consider $\nu = n$ copies [10].

For INDIANA, we support all three approaches (no copies, hardware-oriented model with n copies, and software-oriented model $2n-1$ copies). Internally, however, no changes are made to the circuit model, but the increased leakage effects are simulated by selectively adjusting the leakage probability p of a sampled probe. More precisely, depending on the approach and the fanout of a selected probe, its leakage probability is set to $1-(1-p)^{\nu}$ instead of p.

4 Evaluation

In this section, we evaluate and discuss the performance of INDIANA implementing the verification strategies presented in Section 3. To this end, we verify and benchmark several large-scale cryptographic designs taken from the literature.

We focus our evaluation in particular on the following three aspects. First, we demonstrate that our new verification approach in the glitch-extended (standard) probing model can also be applied to large-scale cryptographic circuits that cannot be analyzed with previous state-of-the-art tools. In particular, we analyze and verify different masking schemes applied to round-based SPN implementations, e.g., PRESENT, SKINNY and AES. Next, we present verification and performance results in the random probing model. To this end, we first focus on an empirical investigation of the parameter choice (i.e., the number of probes and samples) on the accuracy of the leakage probability bounds using an exemplary PRESENT implementation. Finally, we turn to a first-order masked AES implementation, as a case study of greatest practical relevance, and perform random probing checks. In particular, we compare the results with practical measurements that allow us to connect the theoretical models implemented in INDIANA with real implementations.

Verification setup. All reported numbers regarding verification results are derived at a machine equipped with 128 GB RAM and an Intel Xeon E5-1660 Central Processing Unit (CPU) running at 3.2 GHz. The CPU has 16 cores that we fully utilize in case the corresponding tool supports multithreading.

4.1 Glitch-Extended Threshold Probing Verification Results

In this section, we present verification results of single-round implementations of common block ciphers and compare the performance of INDIANA to frameworks from the literature. More precisely, we compare INDIANA to the frameworks maskVerif [3] and VERICA [47]. Before presenting the results, we briefly discuss the reasoning behind this selection and introduce both tools in more detail.

Related Work. Over the last years, many computer-aided formal verification frameworks for analyzing protected hardware implementations have been presented. However, only a few are able to analyze full rounds of protected block ciphers. For example, we tried to execute our case studies with SILVER [34] which, however, is not able to process that many input variables. In this section, we therefore only present our case studies with the maskVerif and VERICA tools that we briefly discuss in the following. In particular, we also exclude PROLEAD from the comparison, as it pursues a non-formal and testing-based approach and for this may exhibit both false positives and false negatives. In contrast to this, maskVerif may only exhibits false negatives (i.e., may classify secure circuits as insecure) while VERICA is exact.

maskVerif. In 2019, Barthe et *al.* presented a novel tool to automatically verify the security of masked implementations in the presence of physical defaults (e.g., glitches and transitions). For that, maskVerif [3] is designed to perform multivariate security analysis, i.e., the tool can analyze probe combinations with probes located at different points in time (e.g., across multiple clock cycles). However, it has been shown that the tool may report false negatives [34]. Indeed, it may falsely categorize a secure circuit as insecure, causing unnecessary design overhead to mitigate such flaws.

VERICA. Richter-Brockmann et *al.* presented the first tool for security verification in the presence of combined attacks, i.e., combining active information tampering with passive information leakage. The tool emerged from the consolidation of SILVER [34] and FIVER [48], a state-of-the-art automated verification tool for FIA. Therefore, VERICA employs BDDs as well for verification of security properties (both for SCA and FIA) [25, 48]. Hence, VERICA can also perform stand-alone security verification in the glitch-extended d-probing model.

Interestingly, VERICA has been implemented as a modular and extendable security verification framework that easily adds additional analysis and verification strategies to the existing tool structure. For this, we opted to integrate our verification concept into this framework to implement INDIANA for verification based on the indistinguishability of probability distributions.

Security Verification. Before we present the verification results in detail, we reason about the selection of target designs and describe how we generate them.

Selection of case studies. Table 2 lists all designs that we consider for our case study. The selected single-round designs cover a range of lightweight ciphers (i.e., PRESENT, ASCON, and SKINNY-64), slightly more complex ciphers (i.e., LED-64 and SKINNY-128), and AES as a widely deployed block cipher. Despite for ASCON and AES, we consider three different protection schemes: Threshold Implementation (TI) [43], HPC1 gadgets [18], and HPC2 gadgets [18]. For the gadget-based designs, we consider configurations protected against first-order and second-order attacks, i.e., $d = 1$ and $d = 2$, respectively.

For the PRESENT and LED implementations protected by TI, we use the same S-box implementation as in [50]. We implemented the remaining (masked) combinatorial logic manually. The TI designs of SKINNY have been designed and implemented by us. The S-boxes for all the HPC1 and HPC2 designs are generated by SAIREDA [23, 24] while the ASCON S-boxes are generated by HADES [15]. Again, we implemented the missing (masked) combinatorial logic manually. The remaining two AES-128 designs were presented in [30] and are based on the Domain-Oriented Masking (DOM) principle.

The implementation costs for the selected case studies are reported in Table 2. We divide the costs into the number of combinatorial gates and the number of memory gates, i.e., registers. Again, these numbers perfectly show the different complexities of the selected designs.

Table 2: Comparison between our indistinguishability strategy and tools from the literature verifying single rounds of various cryptographic schemes. All experiments highlighted in green pass the verification. Experiments highlighted in red fail the verification and are reported as insecure by the corresponding tool. By $\perp$, we denote experiments where the verification did not finish under 8 h. By $\times$, we denote that the computation runs out of memory during parsing or evaluation.

Design	Ord.	Impl.		maskVerif[I] [3]	VERICA[II] [47]	OUR[II]
	d	comb.	mem.			
PRESENT (TI)	1	11 760	192	†	76.5 s	40.8 s
PRESENT (HPC1)	1	1 360	1 152	0.5 s	3.5 s	3.8 s
PRESENT (HPC1)	2	2 816	2 064	3.2 min	$\perp$	55.9 s
PRESENT (HPC2)	1	1 552	1 536	0.5 s^f	4.5 s	5.4 s
PRESENT (HPC2)	2	3 328	3 216	0.9 s^f	$\perp$	11.6 min
LED-64 (TI)	1	12 240	384	†	×	69.3 s
LED-64 (HPC1)	1	1 872	1 152	0.3 s	×	5.5 s
LED-64 (HPC1)	2	3 584	2 064	88.8 s	×	93.9 s
LED-64 (HPC2)	1	2 064	1 536	0.6 s^f	×	8.4 s
LED-64 (HPC2)	2	4 096	3 216	1.1 s^f	×	25.1 min
SKINNY-64 (TI)	1	1 401	192	–*	$\perp$	1.1 s
SKINNY-64 (HPC1)	1	1 064	1 024	–*	9.7 s	2.3 s
SKINNY-64 (HPC1)	2	2 169	1 728	–*	$\perp$	9.1 s
SKINNY-64 (HPC2)	1	1 256	1 408	–*	8.8 s	3.5 s
SKINNY-64 (HPC2)	2	2 873	2 880	–*	$\perp$	39.4 s
SKINNY-128 (TI)	1	1 641	384	–*	$\perp$	2.4 s
SKINNY-128 (HPC1)	1	2 120	3 072	–*	$\perp$	14.0 s
SKINNY-128 (HPC1)	2	4 329	4 992	–*	$\perp$	52.8 s
SKINNY-128 (HPC2)	1	2 504	3 840	–*	$\perp$	21.5 s
SKINNY-128 (HPC2)	2	5 737	7 296	–*	$\perp$	57.4 min
ASCON (HPC1)	1	6 280	4 480	–*	×	67.2 s
ASCON (HPC1)	2	12 424	7 680	–*	×	4.2 h
ASCON (HPC2)	1	7 240	6 400	–*	×	1.7 min
ASCON (HPC2)	2	15 944	13 440	–*	×	×
AES-128 [30]	1	11 312	3 584	4.4 min	×	45.0 min
AES-128 [30]	2	24 184	6 240	$\perp$	×	×
AES-128 (HPC1)	1	11 584	9 088	97.7 s	×	×
AES-128 (HPC2)	1	13 312	12 544	10.5 s^f	×	×

[I] Single-core analysis. [II] Multi-threading (16 cores). † Stack overflow. ^{f} False-negative.
* Round constants cannot be processed by maskVerif.

Verification results. We start our evaluation with the protected implementations of PRESENT. As shown in Table 2, INDIANA is able to verify the security of all five protection schemes. While almost all verifications can be performed in under one minute, the verification of the second-order HPC2 design takes 11.6 min. VERICA is able to analyze the first-order implementations in a similar time range but fails to finish the verification of the second-order designs in under 8 h. For the HPC1 designs, maskVerif is slightly slower for the second-order protected design than INDIANA. However, maskVerif performs a bivariate analysis while INDIANA is limited to intra-cycle verifications. We were not able to perform the verification of the TI design due to a stack overflow. As already mentioned in the introduction of this section, maskVerif may report false negatives. This also happens when verifying the HPC2 designs. We assume that this occurs due to the

computation $a_0 \cdot r + \bar{a_0} \cdot [b_1 + r]$ that is performed inside a HPC2 gadget which is also referred to as the *masked shares multiplication trick*.

Similar results can be observed when analyzing the LED designs. However, VERICA is not able to verify any of these designs because it runs out of memory on our machine.

Next, we verify a number of SKINNY implementations. INDIANA is capable of analyzing all designs while the verification of the SKINNY-128 implementation protected by HPC2 gadgets is the slowest one taking 57.4 min. VERICA only reports results for the first-order HPC designs of SKINNY-64 while performing slightly worse than INDIANA. Eventually, maskVerif is not able to process the SKINNY designs since the tool cannot process round constants.

In our penultimate case study, we analyze the protected ASCON designs. It is evident that these designs exhibit an increased number of combinational and memory gates demonstrating computational limitations of INDIANA on our system. While INDIANA successfully analyzed the HPC1 designs and the first-oder HPC2 design, but was not able to analyze the second-order HPC2 design due to insufficient memory resources.

In our last case study, we perform verifications on AES designs. While VERICA runs out of memory for all designs, INDIANA reports results for the first-order protected implementation from [30]. The verification takes 45 min. Additionally to the first-order design from [30], maskVerif analyses the first-order HPC implementations. As above, it reports false negative results for the HPC2 design.

4.2 Standard Random Probing Verification Results

In this section, we will now focus on the verification and evaluation of different case studies in the standard random probing model in computing intra-cycle random probing leakage probabilities. For the sake of clarity, we will focus on two case studies as examples. However, we would like to note that INDIANA is also able to analyze and verify all designs listed in Table 2 with intra-cycle estimations in the standard random probing model.

Related work. As discussed in the introduction, only a few tools are able to perform verifications in the random probing model. The most recent framework is ironMask, which, however, is limited in the circuit structures that can be parsed and processed, and therefore is not able to analyze entire rounds. Nevertheless, since INDIANA is able to analyze and process the same circuit structures (gadgets) as ironMask, we still offer a detailed comparison in Appendix C, which on the one hand confirms the correctness and on the other hand demonstrates the effectiveness of our proposed approach.

The leakage probability. For all our case studies and experiments, we consider a hypothetical leakage probability $p = 10^{-2}$. However, we would like to note that this choice is not based on practical considerations, but is merely for illustrative purposes, i.e., to provide ranges for the upper and lower limits bounds are visible and displayable. In particular, we would like to emphasize that the actual choice

of p has no influence on the evaluation and verification performance, as this is determined solely by the generation of the leakage function, while the evaluation for different p has the same computational effort.

Case Study I - PRESENT Round Function. Our first case study is therefore dedicated to a first-order HPC2-masked round implementation for the PRESENT cipher, where 16 masked S-boxes are instantiated in parallel (followed by a share-wise bit permutation layer). Moreover, each S-box is composed of PINI-secure HPC2 multiplication gadgets and is evaluated within four clock cycles. Notably, since each of the S-boxes processes an individual part of the secret (i.e., the unshared round input), the verification effort can be divided in *space* (16 parallel instances) and *time* (four clock cycles).

Verification results. In particular, we opted for this case study as we were able to compute exact leakage probabilities for all partitions and cycles of the design, as indicated by all lines in Table 3 marked with ∞ for the number of probes. Based on this, we further opted to investigate various parameters for the maximum number of probes and combinations (denoted as samples) with regard to the accuracy of the leakage probability bounds as part of this experiment. In particular, we carried out various experiments that increased the maximum number of probes and samples in steps of ten and tested all their combinations (with a maximum of 80 probes and 50 samples allowed). Please note that in cases where the number of probes is greater or equal the number of wires of the corresponding cycle, the verification results are the same as performing an exact verification, i.e., as for the first row marked by ∞.

In this context, we would like to highlight and discuss some interesting observations in relation to the results provided in Table 3 and Table 4. However, due to the empirical and incomplete nature of this experiment, we must interpret the results with caution and refrain from making general statements.

Increasing the number of probes. As expected, increasing the number of probes ensures that larger parts of the search space can be covered, which provides more accurate results, especially with regard to the upper bound. It is noteworthy that the lower bound is already quite accurate for most experiments, indicating that the most influential key leakage occurs mainly for smaller combinations of probes.

Increasing the number of samples. Similarly, a larger number of randomly selected samples provides more accurate bounds, although the maximum number of 50 samples was hardly sufficient for any experiment to generate accurate results.

Verification timeouts. Although we were able to generate the exact leak probabilities for all cycles within 23 minutes, we observed timeouts after 8 hours in some experiments. This may seem surprising at first, but can be explained by the caching approach implemented in the CUDD[6] BDD library.

[6] `https://github.com/ivmai/cudd`

Table 3: This table presents verification results for a parameter sweep for a first-order protected implementation of PRESENT (single round) in the random probing model. The table reports the verification results with intra-cycle accuracy for different numbers of probes and samples that are randomly selected by INDIANA. The number of positions corresponds to the total number of wires that is available for probing. By $\perp$, we identify experiments that did not finish in under 8 h.

Cycle	Positions	Probes	Samples				
	part. × wires	part. × probes	16×10	16×20	16×30	16×40	16×50
1	16×15	∞	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$
		16×10	$^{0.012}/_{0.026}$	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$
		16×20	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$
		16×30	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$
		16×40	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$
		16×50	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$
		16×60	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$
		16×70	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$
		16×80	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$	$^{0.013}/_{0.013}$
2	16×58	∞	$^{0.019}/_{0.019}$	$^{0.019}/_{0.019}$	$^{0.019}/_{0.019}$	$^{0.019}/_{0.019}$	$^{0.019}/_{0.019}$
		16×10	$^{0.005}/_{0.908}$	$^{0.009}/_{0.757}$	$^{0.011}/_{0.632}$	$^{0.014}/_{0.565}$	$^{0.015}/_{0.491}$
		16×20	$^{0.014}/_{0.559}$	$^{0.017}/_{0.274}$	$^{0.019}/_{0.158}$	$^{0.019}/_{0.111}$	$^{0.019}/_{0.086}$
		16×30	$^{0.018}/_{0.168}$	$^{0.019}/_{0.050}$	$^{0.019}/_{0.030}$	$^{0.019}/_{0.024}$	$^{0.019}/_{0.022}$
		16×40	$^{0.019}/_{0.031}$	$^{0.019}/_{0.020}$	$^{0.019}/_{0.019}$	$^{0.019}/_{0.019}$	$^{0.019}/_{0.019}$
		16×50	$^{0.019}/_{0.019}$	$^{0.019}/_{0.019}$	$^{0.019}/_{0.019}$	$^{0.019}/_{0.019}$	$\perp$
		16×60	$^{0.019}/_{0.019}$	$^{0.019}/_{0.019}$	$^{0.019}/_{0.019}$	$^{0.019}/_{0.019}$	$^{0.019}/_{0.019}$
		16×70	$^{0.019}/_{0.019}$	$^{0.019}/_{0.019}$	$^{0.019}/_{0.019}$	$^{0.019}/_{0.019}$	$^{0.019}/_{0.019}$
		16×80	$^{0.019}/_{0.019}$	$^{0.019}/_{0.019}$	$^{0.019}/_{0.019}$	$^{0.019}/_{0.019}$	$^{0.019}/_{0.019}$
3	16×82	∞	$^{0.079}/_{0.079}$	$^{0.079}/_{0.079}$	$^{0.079}/_{0.079}$	$^{0.079}/_{0.079}$	$^{0.079}/_{0.079}$
		16×10	$^{0.012}/_{0.995}$	$^{0.016}/_{0.965}$	$^{0.027}/_{0.932}$	$^{0.033}/_{0.904}$	$^{0.039}/_{0.880}$
		16×20	$^{0.034}/_{0.918}$	$^{0.055}/_{0.775}$	$^{0.064}/_{0.654}$	$^{0.069}/_{0.556}$	$^{0.073}/_{0.480}$
		16×30	$^{0.059}/_{0.717}$	$^{0.071}/_{0.447}$	$^{0.076}/_{0.296}$	$^{0.078}/_{0.231}$	$^{0.078}/_{0.191}$
		16×40	$^{0.073}/_{0.400}$	$^{0.078}/_{0.182}$	$^{0.078}/_{0.126}$	$^{0.078}/_{0.103}$	$^{0.078}/_{0.093}$
		16×50	$^{0.078}/_{0.171}$	$^{0.078}/_{0.092}$	$^{0.079}/_{0.082}$	$^{0.079}/_{0.080}$	$\perp$
		16×60	$^{0.078}/_{0.091}$	$^{0.079}/_{0.079}$	$^{0.079}/_{0.079}$	$\perp$	$\perp$
		16×70	$^{0.079}/_{0.079}$	$^{0.079}/_{0.079}$	$\perp$	$\perp$	$\perp$
		16×80	$^{0.079}/_{0.079}$	$^{0.079}/_{0.079}$	$^{0.079}/_{0.079}$	$^{0.079}/_{0.079}$	$^{0.079}/_{0.079}$
4	16×52	∞	$^{0.063}/_{0.063}$	$^{0.063}/_{0.063}$	$^{0.063}/_{0.063}$	$^{0.063}/_{0.063}$	$^{0.063}/_{0.063}$
		16×10	$^{0.020}/_{0.834}$	$^{0.032}/_{0.625}$	$^{0.042}/_{0.515}$	$^{0.048}/_{0.434}$	$^{0.052}/_{0.372}$
		16×20	$^{0.048}/_{0.377}$	$^{0.060}/_{0.188}$	$^{0.062}/_{0.122}$	$^{0.063}/_{0.099}$	$^{0.063}/_{0.087}$
		16×30	$^{0.061}/_{0.120}$	$^{0.063}/_{0.069}$	$^{0.063}/_{0.065}$	$^{0.063}/_{0.064}$	$^{0.063}/_{0.063}$
		16×40	$^{0.063}/_{0.064}$	$^{0.063}/_{0.063}$	$^{0.063}/_{0.063}$	$^{0.063}/_{0.063}$	$^{0.063}/_{0.063}$
		16×50	$^{0.063}/_{0.063}$	$^{0.063}/_{0.063}$	$^{0.063}/_{0.063}$	$^{0.063}/_{0.063}$	$\perp$
		16×60	$^{0.063}/_{0.063}$	$^{0.063}/_{0.063}$	$^{0.063}/_{0.063}$	$\perp$	$\perp$
		16×70	$^{0.063}/_{0.063}$	$^{0.063}/_{0.063}$	$\perp$	$\perp$	$\perp$
		16×80	$^{0.063}/_{0.063}$	$^{0.063}/_{0.063}$	$^{0.063}/_{0.063}$	$^{0.063}/_{0.063}$	$^{0.063}/_{0.063}$

The caching significantly improves performance, especially when calculations are performed repeatedly, i.e., when large parts of the sample combinations overlap. However, if the randomly selected samples are mostly disjoint, this leads to the eviction of cached intermediate results, which ultimately results in slower performance when more samples are considered.

Case Study II - AES Round Function. Our second case study is dedicated to the standardized AES cipher to demonstrate both the power and practical relevance of our approach. In particular, we focus on a first-order DOM-based AES round implementation (i.e., including key addition, 16 parallel S-box instances,

Table 4: This table reports the performance numbers of INDIANA for the verification results presented in Table 3. The performance numbers correspond to the verification time that is required to analyze all four cycles of the target design. By $\perp$, we identify experiments that did not finish in under 8 h.

Probes	**Samples**				
	16×10	16×20	16×30	16×40	16×50
∞	23.0 min	23.0 min	23.1 min	23.0 min	23.1 min
10	6.8 s	6.9 s	7.1 s	7.2 s	7.3 s
20	7.7 s	10.7 s	25.9 s	31.5 s	26.0 s
30	20.4 s	34.6 s	52.4 s	1.1 min	1.4 min
40	51.6 s	1.7 min	2.8 min	5.9 min	11.2 min
50	3.3 min	8.2 min	19.2 min	34.9 min	$\perp$
60	18.3 min	35.5 min	1.3 h	$\perp$	$\perp$
70	49.6 min	2.2 h	$\perp$	$\perp$	$\perp$
80	1.7 h	1.2 h	1.5 h	2.3 h	1.6 h

Table 5: This table presents verification results for a first-order protected implementation of AES (single round) in the random probing model. The table reports the verification results intra-cycle accuracy for combinations of at most 2 probes that are exhaustively generated by INDIANA. The number of positions corresponds to the total number of wires that is available for probing.

Cycle	**Positions**	**Probes**	**Samples**	**Leakage**	**Time**
	part. × wires	part. × probes	part. × samples	min. / max.	totally elapsed
1	16×72	16×2	16×2556	$^{0.056}/_{0.458}$	1.20 min
2	16×138	16×2	16×9453	$^{0.785}/_{0.966}$	6.25 min
3	16×72	16×2	16×2556	$^{0.099}/_{0.472}$	39.33 min
4	16×52	16×2	16×1326	$^{0.145}/_{0.296}$	39.43 min
5	16×52	16×2	16×1326	$^{0.034}/_{0.236}$	39.53 min
6	16×92	16×2	16×4186	$^{0.406}/_{0.738}$	39.79 min
7	16×304	16×2	16×46056	$^{0.992}/_{0.999}$	3.33 h
8	16×102	16×2	16×5151	$^{0.149}/_{0.767}$	3.58 h
9	4×324	16×2	4×52326	$^{0.051}/_{0.981}$	3.76 h

and four parallel MixColumn operations) and provide cycle-accurate ranges for the leakage probabilities in the standard random probing model.

Verification results. Table 5 lists the probes, samples, lower and upper bounds, and total elapsed verification time for the individual clock cycles of the AES round (assuming a leakage probability $p = 10^{-2}$). In particular, due to the complexity of the design (especially for the non-linear stages), we performed our verification with a maximum number of two probes but instead considered all possible combinations (samples). Noting that the design under test is only first-order protected, we assume that the key leakage (i.e., the main contributing failing probe combinations) will be observable with only two probes. However, this still should provide good estimates for the lower bounds while the upper bounds may lack in accuracy due to the large number of unconsidered combi-

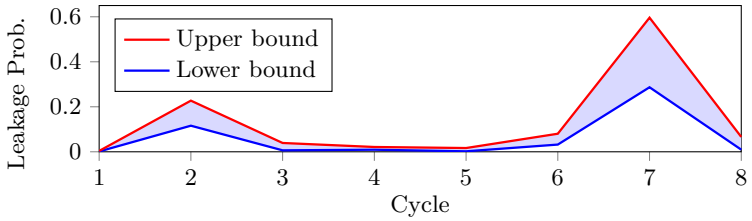


Fig. 1: Verification results of a first-order protected AES implementation (single S-box) in the standard random probing model. The number of probes has been fixed to two while the probe positions have been considered exhaustively.

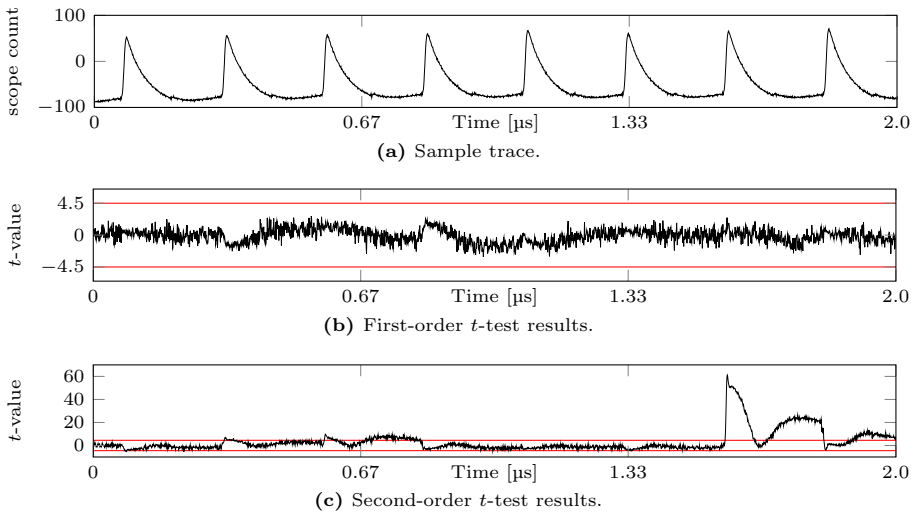
nations (as confirmed by the results in Table 5). Most importantly, the overall verification takes less than 4 hours which is mostly spent in the non-linear layers of cycle 3 and 7.

Case Study III - AES S-Box. For our last case study, we extracted a single S-box instance from the first-order DOM-based AES round function to perform an isolated verification in the standard random probing model alongside empirical TVLA. In particular, the verification results expectedly match with the previous case study. Still, mainly due to the limited capacity of our target Field-Programmable Gate Arrays (FPGAs) platform and in order to reduce the measurement noise for the practical evaluations, we provide this third case study considering an isolated AES S-box instance.

Verification results. As before, we limited the maximum number of samples to two, but took full account of all probe combinations. The results for both the upper bound (red) and the lower bound (blue) are shown in Figure 1.

Practical evaluation. Additionally to the verification results, we perform practical measurements for the same protected AES S-box. Therefore, we synthesize the S-box for the Sakura-G evaluation board that is equipped with a Xilinx Spartan 6 FPGA. The fresh randomness for the AES S-box is provided by a KECCAK core. We set the clock frequency to 4 MHz and measure the power consumption indirectly via a 1 Ω shunt resistor placed in the supply path. The voltage signal is amplified by a ZFL-2000GH+ amplifier and digitized by a Spectrum M4 oscilloscope (8 bit resolution) with a sampling rate of 2.5 GS/s.

The security analysis is performed based on the TVLA methodology originally presented in [28]. Based on Welsh's t-test, we analyze the first two statistical moments with the *fixed vs. random* strategy as described in [49]. Here, the absolute value of the t-test is commonly compared to a threshold of 4.5 corresponding to a confidence of 0.99999 rejecting the null hypothesis. Informally speaking, if we do not identify any point in time that exceeds this threshold, we consider the implementation to be secure. In contrast, if we identify peaks in the test results, the power consumption is not independent of the processed input data such that we cannot conclude that the implementation is secure.



(a) Sample trace.

(b) First-order t-test results.

(c) Second-order t-test results.

Fig. 2: Measurement results for a first-order protected AES S-box (200 Million traces).

Figure 2 shows the measurement results of the protected AES S-box. More precisely, Figure 2a exemplary shows a power trace acquired with the described setup. The eight clock cycles of the S-box can clearly be seen. In total, we collect 200 million power traces (roughly 100 million traces with random input data and 100 million traces with fixed input data). Based on these data, we compute the first-order t-test which is shown in Figure 2b. As expected, we cannot identify any leakage. However, when performing a second-order t-test shown in Figure 2c, we clearly see some peaks indicating the existence of leakage. Particularly, we identify a huge peak in the seventh clock cycle and two smaller peaks in the second and third clock cycle.

Coherency between theory and practice. An interesting observation is that the verification and evaluation results seem to coincide. In both experiments, we observe increased probabilities of leakage detection in clock cycles 2 and 7, with the latter having the highest leakage probability. Additionally, we have limited the experiments to second-order analyses in both cases.

Nevertheless, the results should be considered with caution, as no generally valid conclusions can be drawn from a single experiment. Instead, these observations provide further directions for interesting future work in order to determine and examine the coherency between the theoretical security models and the practical implementations in an in-depth and systematic analysis.

5 Conclusion

In this work, we present INDIANA as a comprehensive verification tool for hardware masking offering verification in the well-established glitch-extended probing

model. Additionally, INDIANA provides intra-cycle estimations for the leakage probabilities in the standard random probing model even for large-scale circuits, e.g., full Substitution-Permutation Network (SPN) ciphers including PRESENT and AES. All this is only possible thanks to our novel and partitionable probing distinguisher and FHT application that enables rapid verification in both models and significantly outperforms state-of-the-art methods based on statistical independence. This is practically demonstrated in providing a comprehensive set of benchmarks and comparisons to state-of-the-art probing security tools.

Acknowledgements

The work described in this paper has been supported by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Germany's Excellence Strategy - EXC 2092 CASA - 390781972, and 510964147 (CAVE). Moreover, this work has been funded in parts by the ERC project 101097056 (SYMTRUST), and by the German Federal Ministry of Education and Research (BMBF) through the projects VE-HEP (16KIS1345), 6GEM (16KISK038), and SASPIT (16KIS1859).

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A Equivalence of Probing Security

In the following, we show the equivalence of our indistinguishability-based definition of probing security, i.e., Definition 4, with the established definition using independence [34]. In particular, the traditional definition requires the statistical independence of the adversarial probe observation from the secret circuit input, i.e., the probability of the probe observation given a specific input is equal to the probability of the probe observation in general. The formal definition is given in Definition 8 where we use the notation used throughout the paper.

Definition 8 (Probing Security - Independence). *Let there be an encoding* $\mathsf{Enc}\colon \mathbb{F}_2^{n_i} \times \mathbb{F}_2^{(s-1)n_i} \to \mathbb{F}_2^{n_i \cdot s}$ *and a circuit* $\mathsf{C} = (\mathcal{G}, \mathcal{W})$ *implementing a function* $\mathsf{F}\colon \mathbb{F}_2^{n_i \cdot s} \times \mathbb{F}_2^{n_r} \to \mathbb{F}_2^{n_o \cdot s}$ *and* $\mathsf{H} \coloneqq \mathsf{A}_\mathsf{C} \circ \mathsf{Enc}$. *We say that* C *is d-probing secure with respect to* Enc *and a probe extension* Ex *if for all* $x \in \mathbb{F}_2^{n_i}$ *and all* $\mathcal{P} \subseteq \mathcal{W}$ *with* $|\mathcal{P}| \leq d$ *it holds that*

$$\mathsf{Pr}_r[\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})}(x,r) = y] = \mathsf{Pr}_{z,r}[\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})}(z,r) = y]$$

Given this, we can show equivalence of this definition to our definition. Here, the intuition is straightforward, if the probe observation is statistically independent of the input then the adversary cannot distinguish different inputs and vice versa.

Theorem 2. *Definition 4 and Definition 8 are equivalent.*

Proof. Let $\mathsf{Enc}\colon \mathbb{F}_2^{n_i} \times \mathbb{F}_2^{(s-1)n_i} \to \mathbb{F}_2^{n_i \cdot s}$ be an encoding and $\mathsf{C} = (\mathcal{G}, \mathcal{W})$ be a circuit implementing a function $\mathsf{F}\colon \mathbb{F}_2^{n_i \cdot s} \times \mathbb{F}_2^{n_r} \to \mathbb{F}_2^{n_o \cdot s}$ and $\mathsf{H} \coloneqq \mathsf{A}_\mathsf{C} \circ \mathsf{Enc}$. Further, let Ex be a probe extension function and $\mathcal{P} \subseteq \mathcal{W}$ be an arbitrary set of probes with $|\mathcal{P}| \leq d$.

$\Rightarrow$ We start with Definition 4 and set $c \coloneqq \mathsf{Pr}_r[\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})}(0,r) = y]$. It holds that

$$\begin{aligned}
\mathsf{Pr}_{z,r}[\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})}(z,r) = y] &= \sum_z \mathsf{Pr}_r[\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})}(z,r) = y]\mathsf{Pr}[z] \\
&= \sum_z c \cdot \mathsf{Pr}[z] \\
&= c \cdot \sum_z \mathsf{Pr}[z] \\
&= \mathsf{Pr}_r[\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})}(x,r) = y]
\end{aligned}$$

and with this Definition 8.

$\Leftarrow$ Now we start with Definition 8, i.e., it holds for all $x \in \mathbb{F}_2^{n_i}$ that

$$\mathsf{Pr}_r[\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})}(x,r) = y] = \mathsf{Pr}_{z,r}[\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})}(z,r) = y]\,.$$

With that, it also holds that

$$\mathsf{Pr}_r[\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})}(0,r) = y] = \mathsf{Pr}_{z,r}[\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})}(z,r) = y]$$

when choosing $x = 0$. Both equations together directly give us Definition 4, which concludes the proof.

B Composability Notions

Unfortunately, the definition of probing security does not imply composability, i.e., the combination of two probing secure circuits is not necessarily probing secure itself. To tackle this problem, different composability notions were introduced, such as Non-Interference (NI) [4], Strong Non-Interference (SNI) [5], and Probe-Isolating Non-Interference (P-INI) [19]. All those have in common that respective circuits can be combined under certain rules to construct larger circuits. The core principle is the restriction of input shares that are statistically dependent on the adversarial observation. Hence, when using indistinguishability instead of independence, we define the leakage not over the unshared secret $x \in \mathbb{F}_2^{n_i}$ but instead over the shared input $\mathsf{Enc}(x) \in \mathbb{F}_2^{n_i \cdot s}$. This means, we assume the specific encoding Enc of masking. Then, we define an index set $\mathcal{I} \subseteq \{0,1,\ldots,n_i \cdot s - 1\}$ that collects all the indices of probe dependent input shares and require the values of the other shares (indicated by the complement set $\overline{\mathcal{I}}$) to be indistinguishable from zero.

Definition 9 (Compositional Leakage). *Let $\mathcal{I} \subseteq \{0,1,\ldots,n_i \cdot s - 1\}$ be an index set and $\mathsf{A_C} \colon \mathbb{F}_2^{n_i \cdot s} \times \mathbb{F}_2^{n_r} \to \mathbb{F}_2^{|\mathcal{W}|}$ be the access to a masked circuit C implementing a function $\mathsf{F} \colon \mathbb{F}_2^{n_i \cdot s} \times \mathbb{F}_2^{n_r} \to \mathbb{F}_2^{n_o \cdot s}$. We define*

$$\mathcal{L}_{(\mathcal{I},\mathsf{A_C},\mathsf{Ex})} \colon 2^{\mathcal{W}} \to \{0,1\}$$
$$\mathcal{P} \mapsto \begin{cases} 0, & \text{if } \forall x_1 \in \mathrm{prec}(\mathcal{I}), x_2 \in \mathrm{prec}(\overline{\mathcal{I}}), y \in \mathbb{F}_2^{|\mathsf{Ex}(\mathcal{P})|} \colon \\ & \mathsf{Pr}_r[\mathsf{A_C}|_{\mathsf{Ex}(\mathcal{P})}(x_1 \oplus x_2, r) = y] = \mathsf{Pr}_r[\mathsf{A_C}|_{\mathsf{Ex}(\mathcal{P})}(x_1, r) = y] \\ 1, & \text{else} \end{cases}$$

to be the compositional leakage of the probe set $\mathcal{P}$ in the masked circuit C, where the probabilities are taken over uniformly random choices of $r \in \mathbb{F}_2^{n_r}$.

As already indicated, there are a lot of different notions of composition proposed in the literature. In this work, we focus on SNI [5], however, emphasize that all other notions can be defined in a similar way using the leakage function from Definition 9. For SNI, we divide the set of probes into d_1 probes on internal wires and d_2 probes on output wires of the circuit, such that the sum is

within the bounds of the security order, i.e., $d_1 + d_2 \leq d$. Then, the set of probe-dependent input shares must be restricted to at most d_1 shares of each input. For this, we assume that the shares belonging to the i'th input are located at the indices $i \cdot s, \ldots, (i+1) \cdot s - 1$. We provide the formal definition in the following:

Definition 10 (Strong Non-Interference - Indistinguishability). *Let* $\mathsf{C} = (\mathcal{G}, \mathcal{W})$ *be a circuit implementing a function* $\mathsf{F}\colon \mathbb{F}_2^{n_i \cdot s} \times \mathbb{F}_2^{n_r} \to \mathbb{F}_2^s$. *We say that* C *is SNI with respect to a probe extension* Ex *if there exists an index set* $\mathcal{I} = \bigcup_{i=0}^{n_i - 1} \mathcal{I}_i$, *with* $\forall i : \mathcal{I}_i \subseteq \{i \cdot s, i \cdot s + 1, \ldots, (i+1) \cdot s - 1\}$ *and* $|\mathcal{I}_i| \leq d_1$, *such that the access* $\mathsf{A_C}$ *to* C *fulfills* $\mathcal{L}_{(\mathcal{I}, \mathsf{A_C}, \mathsf{Ex})}(\mathcal{P}) = 0$ *for all sets* $\mathcal{P} \subseteq \mathcal{W}$ *of* d_1 *internal and* d_2 *output probes, with* $d_1 + d_2 \leq d$.

The proof for equivalence with the traditional notation based on independence [34] follows along the same lines as the respective proof for probing security (cf. Appendix A).

C Comparison to ironMask

In general, INDIANA is able to analyze arbitrary circuits, including small-size gadgets, although our primary focus is to analyze larger designs, i.e., implementations of entire rounds of protected designs. For the sake of completeness and evidence-based confirmation of correctness for INDIANA, however, we still provide additional results on small circuit and compare the performance to ironMask [11].

Table 6: Comparison between INDIANA and ironMask verifying selected gadgets (as reported in the original ironMask literature). By ∞, we denote experiments where the verification did not finish within 1 h.

Gadget		Shares $d+1$	Design wires	ironMask exec. time	INDIANA exec. time
ISW	*refresh*	2	5	0 s	0.0 s
		3	15	0 s	0.0 s
		4	30	0 s	0.0 s
		5	50	75 s	0.0 s
		6	75	∞	0.4 s
		7	105	∞	1.1 s
		8	140	∞	65.6 s
ISW	*addition*	2	14	0 s	0.0 s
		3	36	1 s	0.0 s
		4	68	∞	0.1 s
		5	110	∞	1.0 s
		6	162	∞	14.3 s
		7	224	∞	278.3 s
ISW	*multiplication*	2	21	0 s	0.0 s
		3	57	15 s	0.2 s
		4	110	∞	142.4 s
		5	180	∞	∞

In this context, Table 6 compares the verification performance for INDIANA and ironMask considering two linear (refresh and addition) and one non-linear (multiplication) class of gadgets and various numbers of shares. All verification

results have been obtained on the same verification setup, as outlined in Section 4, however, INDIANA was only able to use a single core (due to the verification of full circuits instead of partitions).

Further, we additionally introduce software-based copy gates to the circuit model in order to reflect the repeated reloading of intermediate values for the random probing model, as it naturally occurs in software implementations. This is in particular necessary, to enable a fair and meaningful comparison to ironMask which only supports such a leakage model. Eventually, we set a time-out of 1 h for all case studies, were ∞ denotes the examples that did not complete within the set period of time.

Interestingly, our approach is competitive and outperforms ironMask for verification of entire (higher-order secure) gadget instances while still allowing to exactly enumerate all failing probe combinations, clearly demonstrating its effectiveness.

D Algorithmic Descriptions

The verification in the glitch-extended threshold probing model is performed according to Algorithm 5.

Algorithm 5: Finding leaking probing sets in the glitch-extended threshold probing model for C with respect to Enc and Ex.

for $\mathcal{P} \subseteq \mathcal{W}$ *with* $|\mathcal{P}| \leq d$ **do**
  Compute $\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})}$ for $\mathcal{P}$
  $|(\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})})_x^{-1}(y)| = \exists_r \prod_{i=0}^{|\mathsf{Ex}(\mathcal{P})|-1} \overline{(\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})_i}(x,r) \oplus y_i)}$
  for $x \in \mathbb{F}_2^{n_i}$, $y \in \mathbb{F}_2^{|\mathsf{Ex}(\mathcal{P})|}$ **do**
    $\mathsf{D}_{\mathsf{Ex}(\mathcal{P})}(y) = |(\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})})_x^{-1}(y)| - |(\mathsf{H}|_{\mathsf{Ex}(\mathcal{P})})_0^{-1}(y)|$
    if $\mathsf{D}_{\mathsf{Ex}(\mathcal{P})}(y) \neq 0$ **then**
      return $\mathcal{P}$
return $\emptyset$

Similarly, the verification in the standard random probing model is performed according to Algorithm 6.

Algorithm 6: Computing the leakage probability ϵ in the standard random probing model for C with respect to Enc and $\mathsf{Ex} = \mathsf{id}$.

$\mathcal{P} \leftarrow \mathcal{W}$
for $x \in \mathbb{F}_2^{n_i}$ **do**
  Compute $\mathsf{D}_{\mathsf{H},x} \colon \mathbb{F}_2^{|\mathcal{W}|} \to \mathbb{R}$
  Compute $\widehat{\mathsf{D}_{\mathsf{H},x}}$ from $\mathsf{D}_{\mathsf{H},x}$ using FFT ▷ `Algorithm 2`
  Compute $\mathcal{L}_{(\mathsf{Enc},\mathsf{A}_\mathsf{C},\mathsf{Ex})}(\mathcal{P})$ from $\widehat{\mathsf{D}_{\mathsf{H},x}}$ ▷ `Algorithm 3`
$\epsilon \leftarrow$ `LeakageProbability`$(\mathcal{L}_{(\mathsf{Enc},\mathsf{A}_\mathsf{C},\mathsf{Ex})}(\mathcal{P}), p)$ ▷ `Algorithm 4`
return ϵ